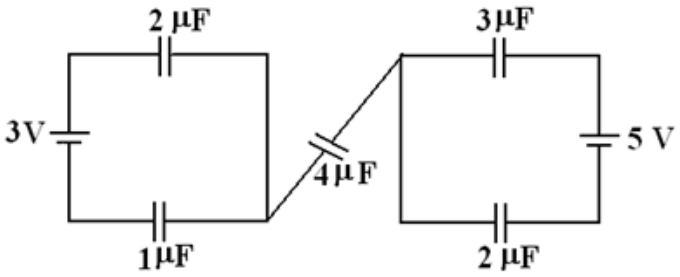
PART-1 : PHYSICS

SECTION-I

1) Which of the following statements is incorrect?

(A) Field obtained by the Superposition of two conservative fields is always conservative.
(B) Field obtained by the Superposition of two non conservative fields is always non-conservative
(C) Field obtained by the Superposition of two non-conservative fields can be non-conservative field
(D) Field obtained by the Superposition of a conservative & non-conservative field will always be non-conservative field

2) All the capacitors were uncharged before they are connected in circuit. Now the charge on 4μF capacitor is



(A) Zero
(B) $4\mu C$
(C) $8\mu C$
(D) $12\mu C$

3) **ASSERTION:** Electric current is a vector quantity
REASON : It has both magnitude & direction

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(B) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(C) Assertion is true but Reason is false
(D) Assertion is false but Reason is true

4) When two identical voltmeters are connected in series to an emf source the readings of each of voltmeter is 6 V. When both of the Voltmeters are connected in parallel to the same emf source their reading is 10 V. Find emf (in Volt).

(A) 8
(B) $\frac{90}{7}$
(C) 12

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(D) 30

5) A body of mass m rises to a height h = R/10 from the surface of earth, where R is the radius of earth. If g is the acceleration due to gravity at the surface of earth the increase in potential energy is

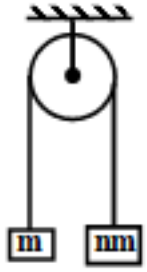
(A) (1/11) mgh

(B) (10/11) mgh

(C) (11/12) mgh

(D) mgh

6) Two blocks of masses m and nm are connected by a massless string passing over a frictionless pulley. The value of n for which both the blocks moves with an acceleration of g/6 is



(A) 5/7

(B) 7/5

(C) Both (A) and (B)

(D) None

7) Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume u_1/u_2 is 1 : 9, the ratio of their diameters d_1/d_2 is

(A) $1 : \sqrt{3}$

(B) 3 : 1

(C) $\sqrt{3} : 1$

(D) 1 : 3

8) Force acting on a particle moving in a straight line varies with the velocity of the particle as $F = \frac{\alpha}{vt}$ where α is constant. The work done by this force from time t_1 to t_2 is :-

(A) $\alpha \ln \frac{t_2}{t_1}$

(B) $\alpha \ln \frac{t_1}{t_2}$

(C) $2\alpha (t_2 - t_1)$

(D) $\alpha^2 (t_2 - t_1)$

9) List-I gives same incomplete nuclear reactions. Match them with the reaction in List-II.

	List-I		**List-II**

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P	$^{237}Np \longrightarrow {}^{233}Pa + ____$	1	β^+ decay
Q	$^{228}Ra \longrightarrow __ + __ + \bar{\nu}$ (antineutrino)	2	β^- decay
R	$^{41}Ca \longrightarrow __ + __ + \nu$ (neutrino)	3	α decay
S	$^{197}_{80}Hg^* \longrightarrow {}^{197}_{80}Hg + __$	4	γ decay

(A) P → 4; Q → 2; R → 1; S → 3

(B) P → 3; Q → 2; R → 1; S → 4

(C) P → 3; Q → 1; R → 2; S → 4

(D) P → 4; Q → 1; R → 2; S → 3

10) A rod of length □ and cross-sectional area A has a variable conductivity given by $K = \alpha + \beta T$, where α, β are a positive constants and T is temperature (in Kelvin) of the isothermal surface. Two ends of the rod are maintained at temperatures T_1 and T_2 ($T_1 > T_2$). Heat current flowing through the rod will be

(A) $\frac{A(T_2-T_1)}{\ell}\left[\beta + \frac{\alpha}{2}(T_2+T_1)\right]$

(B) $\frac{A(T_2-T_1)}{\ell}\left[\beta - \frac{\alpha}{2}(T_2+T_1)\right]$

(C) $\frac{A(T_2-T_1)}{\ell}\left[\alpha - \frac{\beta}{2}(T_2+T_1)\right]$

(D) $\frac{A(T_2-T_1)}{\ell}\left[\alpha + \frac{\beta}{2}(T_2+T_1)\right]$

11) Which of the following phenomenon confirms the transverse nature of light?

(A) Interference

(B) Diffraction

(C) Polarization

(D) Scattering

12) A cubical room of side 5m has a uniform magnetic field B. Total magnetic energy in the room will be.

(A) $\frac{25B^2}{2\mu_0}$

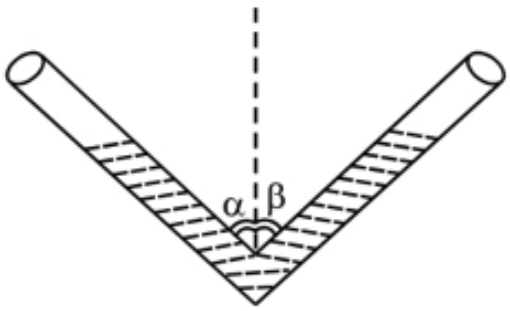
(B) $\frac{125B^2}{2\mu_0}$

(C) $\frac{B^2}{50\mu_0}$

(D) $\frac{B^2}{250\mu_0}$

13) If the liquid is pushed & released, the time period of oscillations will be (length of liquid column = □) (assume the cross sectional area of both sides of the tube to be same)

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(A) $2\pi\sqrt{\frac{\ell}{g(\sin\alpha-\sin\beta)}}$

(B) $2\pi\sqrt{\frac{\ell}{g(\sin\alpha+\sin\beta)}}$

(C) $2\pi\sqrt{\frac{\ell}{g(\cos\alpha+\cos\beta)}}$

(D) $2\pi\sqrt{\frac{\ell}{g(\cos\alpha-\cos\beta)}}$

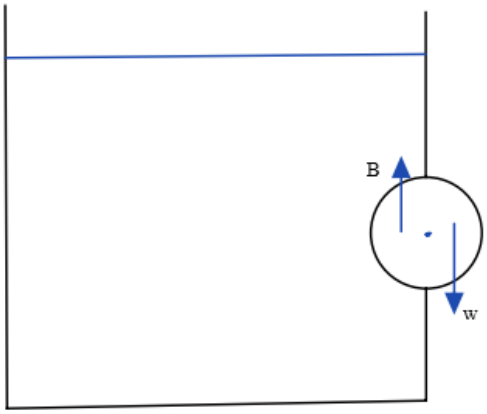
14) A rod of length L has non-uniform linear mass density which varies with distance x from left end of the rod as $\rho(x)=a+b\left(\frac{X}{L}\right)^2$, where a and b are constants and $0 \le x \le L$. The distance of the center of mass of the rod from its left end is :-

(A) $\frac{3}{2}\left(\frac{a+b}{2a+b}\right)L$

(B) $\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)L$

(C) $\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)L$

(D) $\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)L$

15) 

One wall of a large container has a rectangular opening.

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A uniform solid cylinder of density less than that of water is used to plug that rectangular opening as shown. The cylinder is free to rotate about its horizontal fixed axis. The cylinder is kept at rest till the container is filled with water above the top of cylinder as shown and then it is released.
Statement-1 : Just after the cylinder is released it will gain angular acceleration
Statement-2 : The cylinder will acquire zero steady state kinetic energy.

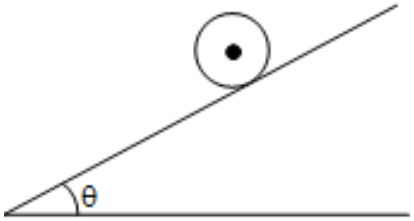
(A) Statement 1 is true, statement -2 is true and statement-2 is correct explanation for statement -1

(B) Statement 1 is true, statement -2 is true and statement-2 is NOT the correct explanation for statement -1

(C) Statement 1 is true, statement -2 is false

(D) Statement 1 is false, statement -2 is true

16) A uniform disc of mass m and radius R is rolling without slipping up a rough inclined plane which makes an angle 30° with the horizontal. If the coefficient of static and kinetic friction are each equal to μ and the only force acting on the disc are gravitational and frictional force then find the direction and magnitude of the frictional force acting on it.



(A) $\frac{mg}{6}$ down the incline

(B) $\frac{mg}{6}$ up the incline

(C) $\frac{mg}{3}$ down the incline

(D) $\frac{mg}{3}$ up the incline

17) A stone projected vertically upwards reaches up to a maximum height h. Assume the potential energy of stone at the point of projection to be zero. The ratio of its Kinetic Energy to its Potential energy at a height $\frac{3}{5}$ h, will be.

(A) 1 : 3

(B) 3 : 1

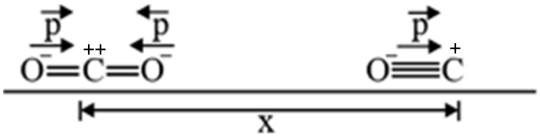
(C) 2 : 3

(D) 3 : 2

18) Carbon dioxide and Carbon monoxide molecules are placed at origin & at a large distance along x-axis respectively as shown. Carbon monoxide can be modelled as a dipole placed along x-axis & Carbon dioxide can be modelled as quadrupole (combination of two dipoles in opposite direction) as shown in diagram. Which of the following statement regarding electrostatic force F between Carbon

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dioxide and Carbon monoxide is most appropriate?



(A) $F \propto \frac{1}{x^4}$

(B) F = 0

(C) $F \propto \frac{1}{x^5}$

(D) force F is attractive in them.

19) When a metallic surface is illuminated with monochromatic light of wavelength λ, the stopping potential is 5 V_0. When the same surface is illuminated with light of wavelength 3λ, the stopping potential is V_0. Then the work function of the metallic surface is :

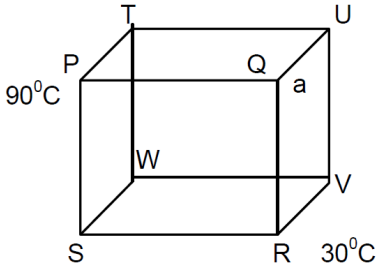
(A) $\frac{hc}{6\lambda}$

(B) $\frac{hc}{5\lambda}$

(C) $\frac{hc}{4\lambda}$

(D) $\frac{2hc}{4\lambda}$

20) 12 identical rods made of same material are arranged in the form of a cube. The temperature of 'P' and 'R' are maintained at 90° C and 30° C respectively. Then the temperature of point 'V', when steady state is reached is :-



(A) 65°C

(B) 60°C

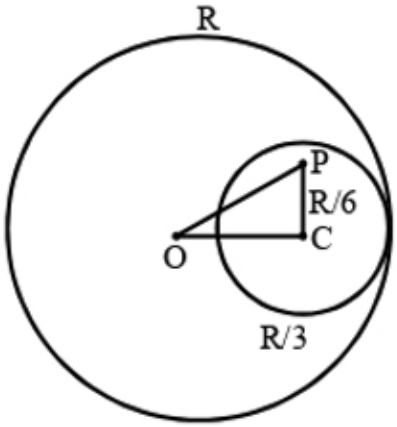
(C) 20°C

(D) 50°C

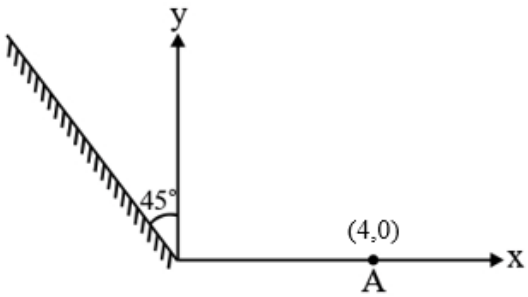
SECTION-II

1) A solid uniformly charged non-conducting sphere of radius R has a spherical cavity of radius $\frac{R}{3}$ as shown. The ratio of magnitudes of electric field at center of Cavity and that at a distance $\frac{R}{6}$ from

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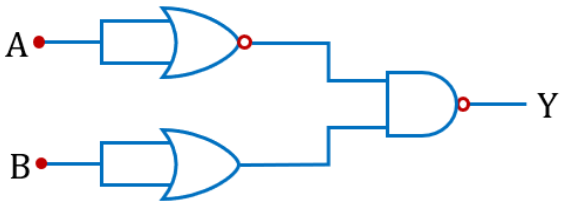
center of Cavity is

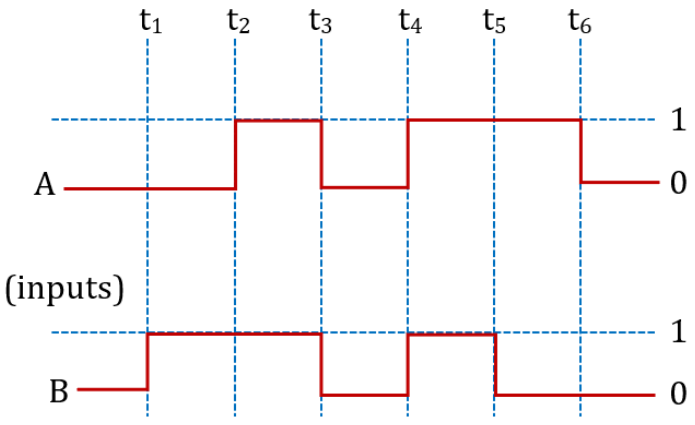


2) What will be magnitude of y - coordinate of image of point A?

3) A stone is projected horizontally at a speed of 20 m/s from the top of a high tower. Simultaneously another stone is released from the same point. Find the distance in meters between stones after 1.4 seconds

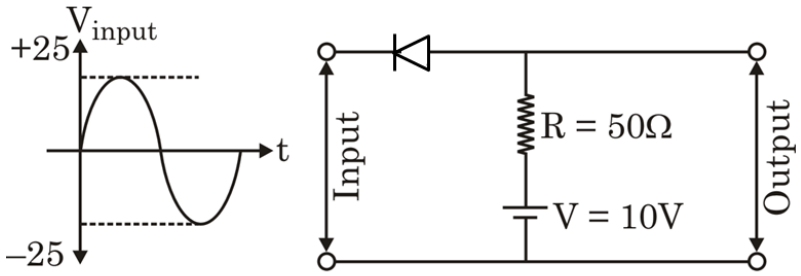
4) For the following circuit and given inputs A and B, find output between t_5 to t_6.





5) An ideal diode is connected in a circuit with resistance R = 50Ω and V = 10 volt as shown in figure, maximum value of output voltage (in V) in the positive cycle will be :

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PART-2 : CHEMISTRY

SECTION-I

1) 1 mole of borax reacts with how many mole of H_2SO_4 ?

(A) 1
(B) 2
(C) 4
(D) Reaction does not take place

2) In which of the following C-H bond has the highest 's' character ?

(A) acetylene
(B) ethylene
(C) methane
(D) Benzene

3) **Assertion :** The complex $[CoCl_2(en)_2]^+$ has an octahedral geometry
Reason : The complex has noble gas E.A.N

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(B) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(C) Assertion is true but Reason is false.
(D) Assertion is false but Reason is true.

4) The average value of C–C bond order in graphite is

(A) 1
(B) 3/2
(C) 3/4
(D) 4/3

5) Which of the following specie has the bond order of 0.5

(A) C_2^+

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(B) B_2^+

(C) N_2^+

(D) O_2^+

6) Which of the following compound is formed in borax bead test ?

(A) Orthoborate

(B) Metaborate

(C) Double oxide

(D) Tetraborate

7) Weight ratio of Fe : C in $Fe_2[Fe(CN)_6]$ is : [Fe = 56]

(A) 3/7

(B) 7/5

(C) 7/3

(D) 5/7

8) The degree of dissociation of HI at a particular temperature is 0.8. The volume of 2M hypo solution required to react completely with the iodine present in an equilibrium mixture of reaction, which is started by 2 mole each of H_2 and I_2 in a vessel of two litre capacity, is

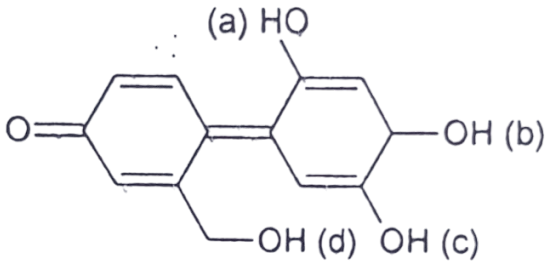
(A) 0.4 litre

(B) 0.8 litre

(C) 1.6 litre

(D) 3.2 litre

9) Most acidic hydrogen among the following is :



(A) a

(B) b

(C) c

(D) d

10)

The moles of H^+ from H_2O alone in a $1l,\ \sqrt{5} \times 10^{-7}$ M HCl solution at 25°C is $(\sqrt{5} = 2.23)$

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(A) 10^{-7}

(B) 6.85×10^{-8}

(C) 3.85×10^{-8}

(D) 10^{-8}

11) 5 moles of an ideal gas undergoes isothermal expansion from 4.0 bar to 1.0 bar, at 27°C. The minimum amount of heat absorbed by the gas is (ln2 = 0.70)

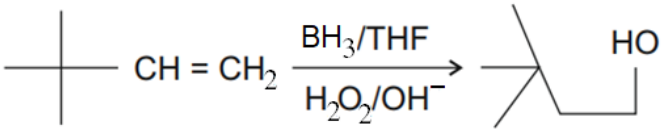
(A) 2250 cal

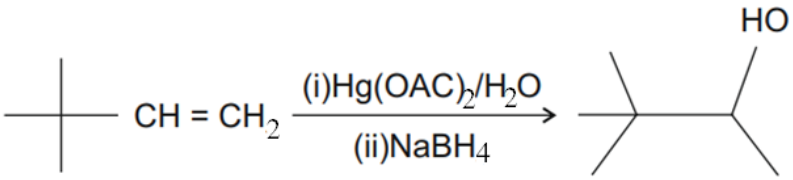
(B) 4200 cal

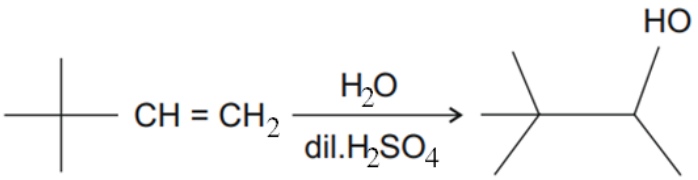
(C) 4500 cal

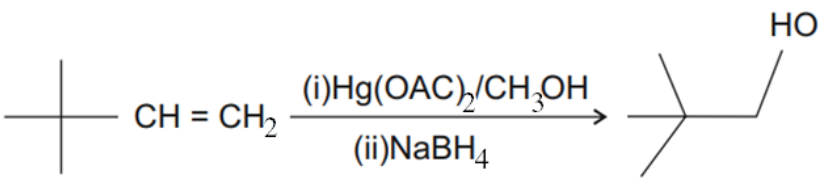
(D) 450 cal

12) Alkenes form alcohols by hydration reactions in which water is added in presence of dil. H_2SO_4. Addition of water follows Markownikoff's rule. For unsymmetrical alkenes, more than one product is formed. The major product is decided by stability of the intermediate carbocation which is formed by addition of H^+ . Alcohol formation is also done by treating alkenes with other reagents.
Answer the following question on the basis of above writer up.

(I) 

(II) 

(III) 

(IV) 

In which reactions, the product formed is correctly given?

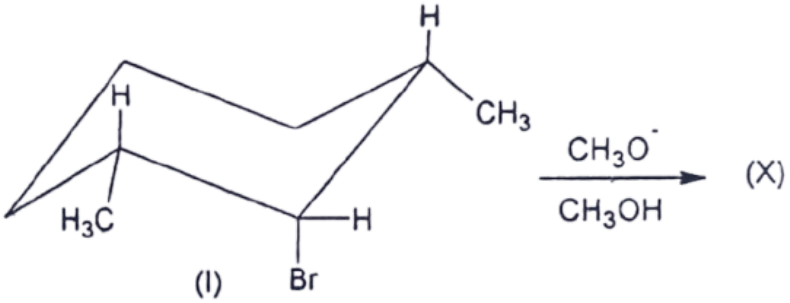
(A) I, II, III

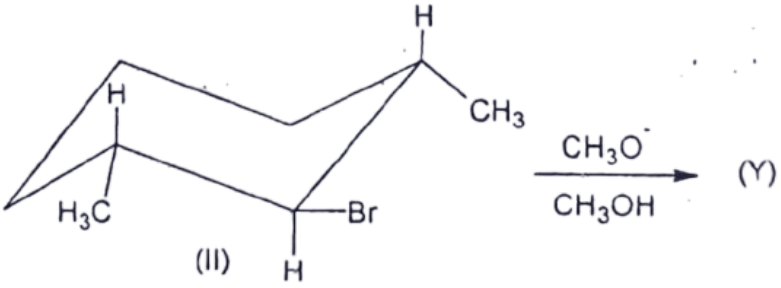
(B) I, II

(C) I, III, IV

(D) III, IV

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13)
Which of the following statement is NOT correct

(A) Compound (I) readily undergoes elimination than compound (II)

(B) Compound (II) readily undergoes elimination than compound (I)

(C) Elimination in (I) and (II) easily takes place when eliminating groups are anti periplanar to each other

(D) Product (X) forms via E_2 elimination

14) The values of conductance, specific conductance, equivalent conductance and molar conductance of a definite volume of an electrolyte at a particular temperature are A mho, B S cm^{-1}, C S cm^{-2} $equiv^{-1}$ and D S $cm^{-2}mole^{-1}$ respectively. Which of the following will decrease on dilution?

(A) D

(B) B

(C) C

(D) Only C and D

15) Match the isoelectric points with amino acids:

Amino acids	Isoelectric points
(W) H_2N - CH_2COOH	(I) 9.5
(X) HOOC - $CH_2CH_2CH(NH_2)COOH$	(II) 6.0
(Y) $H_2N(CH_2)_4CH(NH_2)COOH$	(III) 3.1
(Z) $H_2NCOCH_2CH(NH_2)COOH$	(IV) 5.6

(A) W - II, X - III, Y - I, Z - IV

(B) W - III, X - I, Y - IV, Z - II

(C) W - III, X - I, Y - II, Z - IV

(D) W - II, X - I, Y - III, Z - IV

16) The solubility (expressed as mole fraction of gas dissolved in final solution) of any gas in any liquid increases by :

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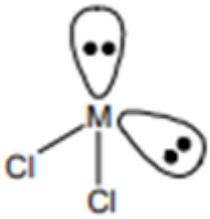
(A) Increase in temperature at constant pressure

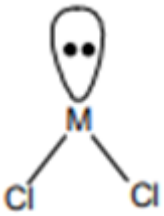
(B) Increase in amount of solvent at same temperature and pressure

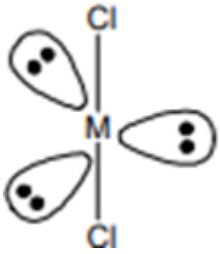
(C) Increase in pressure at constant temperature

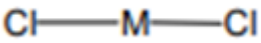
(D) Increase in moles of gas above solution keeping pressure and temperature constant.

17) If the atomic number of M is 50. The shape of gaseous MCl_2 is

(A) 

(B) 

(C) 

(D) 

18) When N_2 goes to N_2^+, the $N-N$ bond distance ..... and when O_2 goes to O_2^+ the $O-O$ bond distance....

(A) increases, decreases

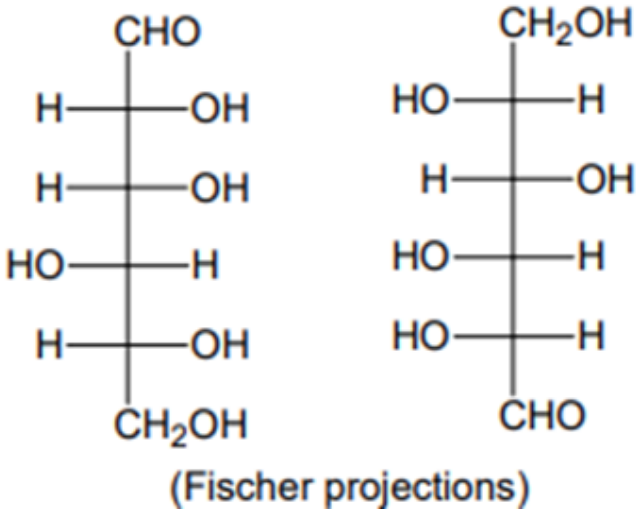
(B) decreases, increases

(C) increase in both case

(D) decrease in both case

19) What is the relationship between the molecules in the following pairs?

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(Fischer projections)

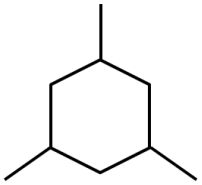
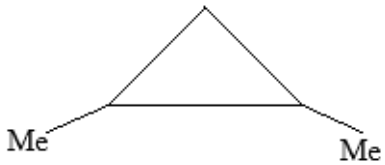
(A) enantiomers

(B) diastereomers

(C) identical

(D) structural isomers

20)

Column-I		Column-II	
(A)	$CH_3 - CH = CH - CH = N - OH$	(P)	2
(B)		(Q)	4
(C)		(R)	3
(D)	$CH_3 - CH = CH - CH = CH - CH(CH_3) - Ph$	(S)	8

Match stereo isomers for above compounds

(A) A $\rightarrow$ Q, B $\rightarrow$ P, C $\rightarrow$ R, D $\rightarrow$ S

(B) A $\rightarrow$ S, B $\rightarrow$ P, C $\rightarrow$ R, D $\rightarrow$ Q

(C) A $\rightarrow$ Q, B $\rightarrow$ R, C $\rightarrow$ P, D $\rightarrow$ S

(D) A $\rightarrow$ P, B $\rightarrow$ P, C $\rightarrow$ Q, D $\rightarrow$ S

SECTION-II

1) The total number of diprotic acids among the following is

H_3PO_4 H_2SO_4 H_3PO_3 H_2CO_3 $H_2S_2O_7$

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H_3BO_3 H_3PO_2 H_2CrO_4 H_2SO_3

2) For a second order reaction, containing single reactant : A → B, it is given that :
$T_{75\%} = X \cdot t_{50\%}$
Find the value of X

3) The molecular mass of an aliphatic ketone is 86. It forms oxime with NH_2OH. The oxime is reduced to an resolved amine. How many structural isomer (s) of the ketone form (s) resolved amine?

4) $CH_2=C=C=C=CH_2$
How many maximum number atoms of the above compounds are present in one plane?

5) Total number of stereoisomerism present in complex compound [Cr(gly)(NH_3)(H_2O)C□Br].

PART-3 : MATHEMATICS

SECTION-I

1) Let $f(x) = \int \frac{x^2 dx}{(1+x^2)\left(1+\sqrt{1+x^2}\right)}$ and f (0) = 0. Then f (1) is equal to

(A) $\ln\left(1+\sqrt{2}\right)$

(B) $\ln\left(1+\sqrt{2}\right) - \frac{\pi}{4}$

(C) $\ln\left(1+\sqrt{2}\right) + \frac{\pi}{4}$

(D) None of these

2) A is 3 x 3 matrix whose entries belong to the set $\{-1, 0, -1\}$. Then the probability that A is neither symmetric nor skew-symmetric is

(A) $\frac{3^9 - 3^6 - 3^3 + 1}{3^9}$

(B) $\frac{3^9 - 3^6 + 3^3 + 1}{3^9}$

(C) $\frac{3^9 - 3^6 - 3^3}{3^9}$

(D) $\frac{3^9 - 3^6 + 3^3}{3^9}$

3) Consider the function $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) + 2\tan^{-1}x$

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Assertion (A) : $f(10) = \pi$
Reason (R) : $f(x) = \pi \; \forall \; x \geqslant 1$

(A) **A** is true, **R** is true and **R** is correct explanation for **A**

(B) **A** is true, **R** is true and **R** is NOT the correct explanation for **A**

(C) **A** is true, **R** is false

(D) **A** is false, **R** is true

4) If z is a complex number which satisfies the equation $|z-5-7i| = 9$, then the maximum value of $|z-2-3i|$ is equal to

(A) 4

(B) 9

(C) 12

(D) 14

5) A pair of mutually perpendicular lines is drawn from origin to form an isosceles triangle with the line $2x+3y=6$. The area of triangle thus formed is equal to

(A) $\frac{36}{13}$

(B) $\frac{12}{17}$

(C) $\frac{13}{5}$

(D) $\frac{6}{13}$

6) The curve passing through (0, 1) which satisfies the differential equation $\frac{xdy-ydx}{xdy+ydx} = y^2 \sin(xy)$, is given by

(A) $x+y(1+\cos(xy))=0$

(B) $x-y(1-\cos(xy))=0$

(C) $x+y(1-\cos(xy))=0$

(D) $x-y(1+\cos(xy))=0$

7) A latus rectum subtends a right angle at the other focus of the hyperbola. The eccentricity of the hyperbola must be

(A) $1-\sqrt{2}$

(B) $\sqrt{2}$

(C) $\sqrt{2}-1$

(D) $1+\sqrt{2}$

8) The first term of an Arithmetic Progression (AP) with real terms is -1 and the 8th term of a Geometric Progression (GP) with real terms is 80. The 4th terms of both progressions are equal. If

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the 6th term of the GP is equal to the sum of the 6th and 7th terms of the AP, then the smallest possible integral value of the 9th term of the GP is equal to

(A) 160

(B) 2

(C) -160

(D) -2

9) Consider the following lists

	List-I		List-II
(A)	$\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right] : \sin x + \cos x = 1\right\}$	(I)	Has elements 2
(B)	$\left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}$	(II)	Has elements 3
(C)	$\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$	(III)	Has elements 4
(D)	$\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$	(IV)	Has elements 6

(A) $A \to I; B \to IV; C \to I; D \to IV$

(B) $A \to I; B \to I; C \to IV; D \to III$

(C) $A \to II; B \to I; C \to IV; D \to IV$

(D) $A \to II; B \to IV; C \to I; D \to III$

10) The value of $\int_0^{\infty} \left[\frac{3}{1+x^2}\right] dx$ ([.] denotes greatest integer function) is

(A) 3

(B) $\frac{3}{\sqrt{2}}$

(C) $3\sqrt{2}$

(D) infinite

11) The area of the region A represented by $A = \{(x, y) \mid (x, y) \in R \times R,\ 1 \leqslant |x-2| + |y+1| \leqslant 2\}$ is

(A) 9 sq. units

(B) 4 sq. units

(C) 5 sq. units

(D) 6 sq. units

12) If the line $y = 2 - x$ is tangent to the circle S at a point P (1,1) and circle S is orthogonal to the circle $x^2 + y^2 + 2x + 2y - 2 = 0$, then the equation of circle S is

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(A) $x^2 + y^2 = 2$

(B) $x^2 + y^2 - 3x - 3y + 4 = 0$

(C) $x^2 + y^2 + x - 3y = 0$

(D) $x^2 + y^2 - x - y = 0$

13) A fair dice is thrown three times and the respective outcomes are named a, b and c. The probability that $f(x) = x^3 + ax^2 + bx + c$ is a increasing function is

(A) $\frac{5}{36}$

(B) $\frac{4}{9}$

(C) $\frac{1}{3}$

(D) None of these

14) A focal chord of the parabola $y^2 = 16x$ is a tangent to the circle $(x-6)^2 + y^2 = 2$. The length of this focal chord is equal to

(A) 32

(B) 16

(C) 512

(D) 64

15) Consider the following statements about the function $f : A \to A$
Statement-1 : If f is onto, then it must be one-one.
Statement-2 : If f is one - one, then it must be onto.

(A) Only (S1) is true

(B) Only (S2) is true

(C) Both (S1) and (S2) is true

(D) Both (S1) and (S2) is false

16) An ellipse having foci (3,1) and (1,1) and passing through the point (1,3) has eccentricity equal to

(A) $\sqrt{2}-1$

(B) $\frac{\sqrt{2}-1}{2}$

(C) $\sqrt{3}-1$

(D) $\frac{\sqrt{3}-1}{2}$

17) Let the observations $x_i (1 \le i \le 10)$ satisfy the equations, $\sum_{i=1}^{10}(x_i - 5) = 10$ and $\sum_{i=1}^{10}(x_i - 5)^2 = 40$.
If μ and λ are the mean and the variance of the observations, $x_1 - 3$, $x_2 - 3$, ..., $x_{10} - 3$, then the

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ordered pair (μ, λ) is equal to :

(A) (6, 6)

(B) (3, 6)

(C) (6, 3)

(D) (3, 3)

18) The value of $\tan 70^{\circ} - \tan 50^{\circ} + \tan 10^{\circ}$ equals

(A) 0

(B) 2

(C) 3

(D) $\sqrt{3}$

19) Suppose $p, q, r \neq 0$ and the system of equations

$$(p+a)x + by + cz = 0$$
$$ax + (q+b)y + cz = 0$$
$$ax + by + (r+c)z = 0$$

has non-trivial solutions, then the value of $\frac{a}{p} + \frac{b}{q} + \frac{c}{r}$ is

(A) -1

(B) 0

(C) 1

(D) 2

20) If p, q, r and s are non-zero numbers such that r, s are roots of $x^2 + px + q = 0$ and p, q are the roots of $x^2 + rx + s = 0$, then the value of $(4p + q + 2r + s)$ is

(A) 0

(B) 1

(C) 2

(D) 4

SECTION-II

1) The value of $\lim\limits_{x \to 2} \frac{2^x + 2^{3-x} - 6}{2^{-\frac{x}{2}} - 2^{1-x}}$ is

2) If $\vec{a}$ and $\vec{b}$ are any two unit vectors, then the maximum possible integral value of $3\left|\vec{a} - \vec{b}\right| + \frac{7}{2}\left|\vec{a} + \vec{b}\right|$ is equal to

3) The shortest distance between the lines $\frac{x-1}{2} = \frac{y}{3} = \frac{z}{2}$ and $\frac{x-3}{2} = \frac{y}{5} = \frac{z-2}{4}$ is equal to

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4) The remainder when 5^{99} is divided by 13 is

5) How many words with or without meaning formed using all the letters of the word ALLEN begin and ends with a consonant ?

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ANSWER KEYS

PART-1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	B	A	D	B	B	C	A	A	B	D	C	B	C	C	D	B	C	C	A	D

SECTION-II

Q.	21	22	23	24	25
A.	1	4	28	1	10

PART-2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	A	A	B	D	B	B	C	C	C	C	A	B	B	B	A	C	B	A	C	A

SECTION-II

Q.	46	47	48	49	50
A.	6	3	2	7	24

PART-3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	B	A	A	D	A	C	D	C	B	B	D	D	B	A	D	A	D	D	A	C

SECTION-II

Q.	71	72	73	74	75
A.	8	9	2	8	18

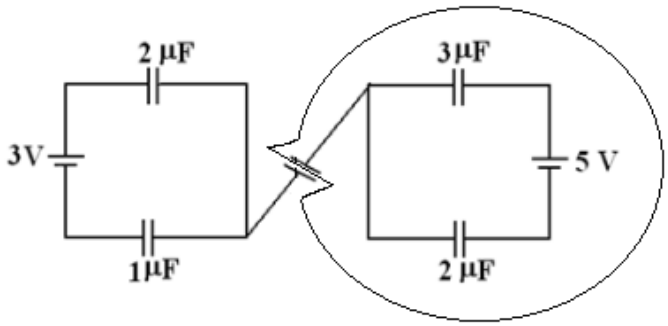
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SOLUTIONS

PART-1 : PHYSICS

1)

Conceptual.

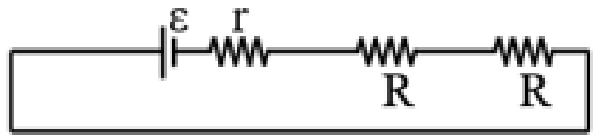


2)
Net charge on this isolated system shown continuous to be zero. The both plates of any capacitor contain zero charge. The middle capacitor has only one plate inside the isolated system taken as shown in diagram and therefore its single plate contains zero charge.

3)

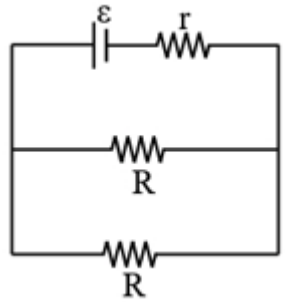
Electric current does not follow laws of vector edition and hence it is not a vector quantity

4)



$$\frac{ER}{r+2R} = 6.....1$$

$$\frac{E\frac{R}{2}}{(r+\frac{R}{2})} = 10$$



$$\frac{ER}{2r+R} = 10.....2$$

$$\frac{2r+R}{r+2R} = \frac{6}{10}$$

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$10r + 5R = 3r + 6R$
$7r = R = \ldots\ldots\ldots 3$
using.....1
$\frac{E7r}{15r} = 6$
$E = \frac{90}{7}$v.....ans

5)

$$\Delta U = \left(\frac{GMn}{\frac{11R}{10}}\right) - \left(\frac{GMn}{R}\right)$$

6) Case--1 :
T-mg = mg/6.....1
nmg-T = nmg/6.....2
Solving (1) & (2) we get
n = 7/5

Case--2:
mg - T = mg/6.....1
T - nmg = nmg/6.....2
Solving (1) & (2) we get
5/7

8) $W = \int Fvdt$

10) $\frac{dQ}{dt} = -\frac{KAd\theta}{dx}$

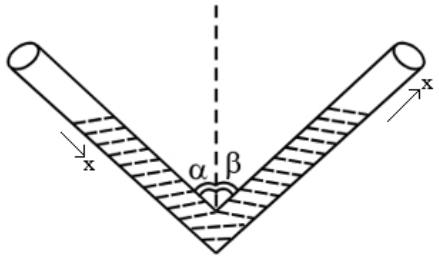
11)

Polarization takes place only in transverse waves and not in longitudinal waves.

12)

Magnetic energy density = $\frac{B^2}{2\mu_0}$

13)

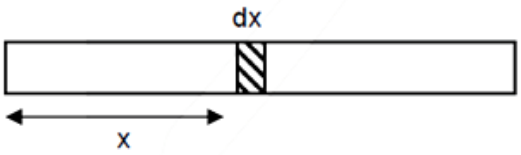


$F = \Delta P.a = -\rho gA(x\cos\alpha + x\cos\beta)$
$a = -\frac{\rho gA(\cos\alpha + \cos\beta)}{m}x = \frac{\rho gA(\cos\alpha + \cos\beta)}{\ell.A.\rho}x$
$= -\omega^2 x$

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$$\omega = \sqrt{\frac{g(\cos\alpha + \cos\beta)}{\ell}} \Rightarrow T = 2\pi\sqrt{\frac{\ell}{g(\cos\alpha + \cos\beta)}}$$

14)



$$X_{cm} = \frac{1}{M}\int_0^{\ell} X.dM$$

$$dM = \rho.dx = \left(a + b\left(\frac{X}{L}\right)^2\right).dx$$

$$x_{cm} = \frac{\int x dM}{\int dM} = \frac{\int \rho dx}{\rho dx} = \frac{\int_0^L x\left(a + \frac{bx^2}{L^2}\right)dx}{\int_0^L \left(a + \frac{bx^2}{L^2}\right)dx}$$

$$= \frac{a\left(\frac{x^2}{2}\right)_0^L + \frac{b}{L^2}\left(\frac{x^4}{4}\right)_0^L}{a(x)_0^L + \frac{b}{L^2}\left(\frac{x^3}{3}\right)_0^L} = \frac{(2a+b)L}{(3a+b)a} \times 3$$

16) For disk :

$mg \sin\theta - f_r = m(a)$

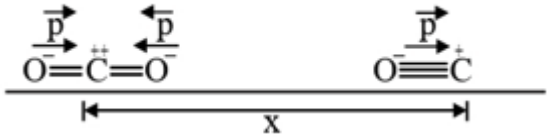
& $(f_r)R = I(\alpha) = \left(\frac{MR^2}{2}\right)\frac{a}{R}$

$\Rightarrow f_r = \frac{mg}{6}$

17) $E_k = \frac{1}{2}mv^2 = \frac{2mgh}{5}$

pot. energy = $\frac{3mgh}{5}$

18)



$\vec{E}$ due to dipole $\propto \frac{1}{x^3}$

$\vec{E}$ due to quadmpole $\propto \frac{1}{x^4}$

$\therefore \vec{E}$ due to CO_2 molecules $\propto \frac{1}{x^4}$

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force in dipole placed in that field $\propto \frac{1}{x^5}$

19) $\frac{hc}{\lambda} = 5\ eV^0 + \phi$

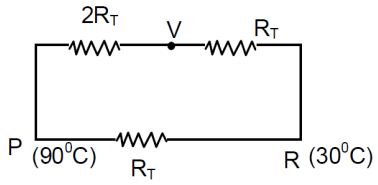
$\frac{hc}{3\lambda} = eV^0 + \phi$

$\Rightarrow \frac{2hc}{3\lambda} = 4eV^0$

$\Rightarrow \phi = \frac{hc}{6\lambda}$

20) Equivalent circuit is

$\frac{90-V}{2R_T} = \frac{20}{R_T} \Rightarrow V = 50°C$



21)

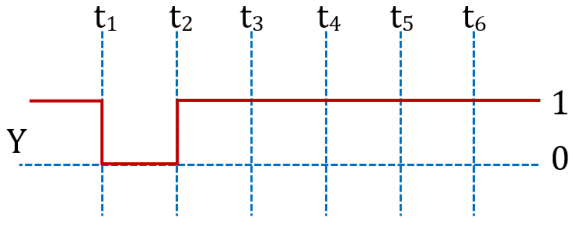
Magnitude of Electric field at both P and C is $\rho\frac{\overrightarrow{OC}}{3\varepsilon_0}$

22)

The image of +ve x-axis lies along -ve y-axis therefore coordinate of image is (0,-4)
Magnitude of y-coordinate of image point = 4.....Ans.

23) As relative acceleration of stones is 0 the relative motion will be uniform motion
S = vt
= 20 X 1.4
= 28

24) $Y = \overline{\overline{A}\cdot B} = A + \overline{B}$



25) Positive half cycle

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V_{output} = 10V

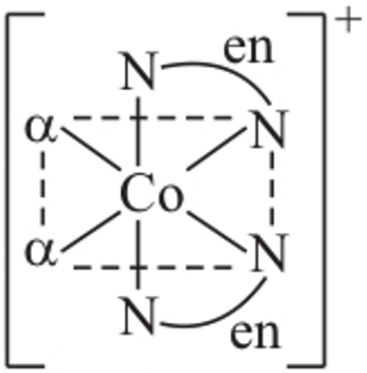
PART-2 : CHEMISTRY

26)

Ans. (A)
1 mole of borax reacts with H_2SO_4
$Na_2B_4O_7.10H_2O + H_2SO_4 \rightarrow 4H_3BO_3 + Na_2SO_4 + 5H_2O$

27)

Ans. (A)
Solution $\rightarrow$ In acetylene, $CH \equiv CH$ the C is sp hybridised with 50% character.



Octahedral
(cis)

28)
$[CoCl_2(en)_2]^+$
$Co^{3+} \rightarrow 24$
EAN = 24 + 2(2) + 2(4) = 36
36 is atomic no. of Kr.

29)

The average value of C-C bond order in graphite is 1.33

30)

Specie	B.O.
C_2^+	1.5
B_2^+	0.5
N_2^+	2.5
O_2^+	2.5

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31)

Borax + CuO → $2NaBO_2 + Cu(BO_2)_2 + H_2O$) blue metaborate

32)

mass of Fe in $Fe_2[Fe_2(CN)_6]$ is 168
mass of C in $Fe_2[Fe_2(CN)_6]$ is 72
Weight ratio of Fe:C is $\frac{7}{3}$

33)

Keq = 4
$2Na_2S_2O_3 + I_2 \rightarrow 2NaI + Na_2S_4O_6$

34)

Most stable extented conjugation.

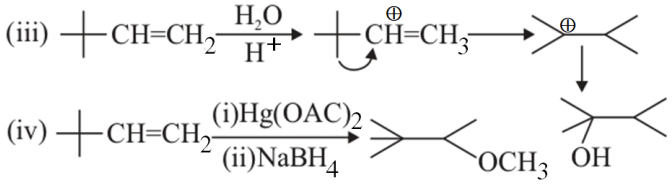
35)

$HCl \rightarrow H^+ + Cl^-$
$\left(\sqrt{5} \times 10^{-7} + x\right)$
$H_2O \rightarrow H^+ + HO^-$
$\sqrt{5}\times10^{-7}+x \quad x$
$x \times \left(x + \sqrt{5} \times 10^{-7}\right) = 10^{-14}$
$x^2 + \sqrt{5}x \times 10^{-7} - 10^{-14} = 0$
$x = \frac{-\sqrt{5} \times 10^{-7} + \sqrt{5 \times 10^{-14} + 4 \times 10^{-14}}}{2}$
$= \frac{-\sqrt{5} \times 10^{-7} + \sqrt{9 \times 10^{-14}}}{2} = \frac{3 - \sqrt{5}}{2} \times 10^{-7}$
$x = 3.85 \times 10^{-8}$

36) For minimum heat absorbed, process will be single step irreversible.
$q = \Delta U - W = 0 - [-P_{ext}(V_2.V_1)]$
$= P_2\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right)$
$= nRT\left(1 - \frac{P_2}{P_1}\right)$
$= 5 \times 2 \times 300\left(1 - \frac{1}{4}\right) = 2250$ cal.

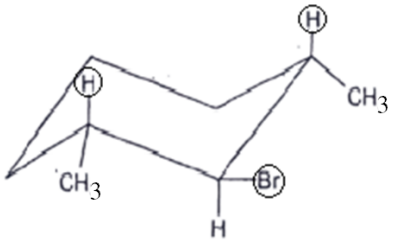
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37)



38)

In (II) compound the eliminating groups are not anti-periplanar to each other



39)

Specific conductance decreases on dilution

40)

Ans. (A)
W - Neutral Amino Acid
X - Acidic Amino Acid
Y - Basic Amino Acid
Z - Neutral Amino Acid

41) Theory based (Henry's law)

42)

Ans. (B)
Atomic no. 50 is Sn
gaseous $SnCl_2$ is bent.

43)

Ans (A)

Bond distance $\propto \frac{1}{\text{Bondorder}}$

BO of N_2 is 3
BO of N_2 is 2.5
BO of O_2 is 2
BO of O_2^+ is 2.5

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44)

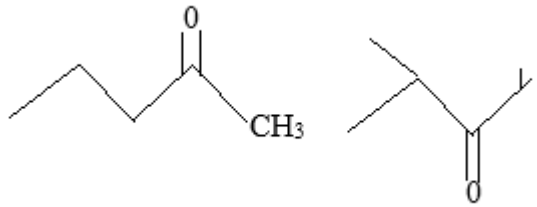
Ans (C)
they are identical

46) Ans (6)
H_2SO_4, H_3PO_3, H_2CO_3, H_2S_2O7, H_2CrO_4, H_2SO_3

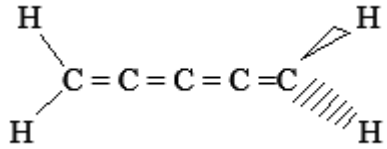
47) Ans 3.

$\frac{I}{At} - \frac{1}{A_0} = Kt,$ $\frac{3}{AO} = Kt75\%$ $\frac{1}{AO} = Kt50\%$

48) Ans. 2



49) Ans. 0



PART-3 : MATHEMATICS

51) Let $\tan^{-1} x = \theta$. Then $x = \tan\theta$ and $dx = \sec^2\theta d\theta$

$$f(\tan\theta) = \int \frac{\tan^2\theta.\sec^2\theta}{\left(1+\tan^2\theta\right)(1+\sec\theta)}d\theta = \int \frac{\tan^2\theta}{(1+\sec\theta)}d\theta = \int \frac{\left(\sec^2\theta-1\right)}{(1+\sec\theta)}d\theta$$

$$= \int (\sec\theta - 1)\,d\theta = \ln|\sec\theta+\tan\theta| - \theta + c$$

Thus, $f(0) = 0 \Rightarrow c = 0$

Therefore, $f(1) = \ln\left(1+\sqrt{2}\right) - \frac{\pi}{4}$

52) Total number of matrices = 3^9
Number of symmetric matrices = 3^6
Number of skew-symmetric matrices = 3^3
Since, Null matrix is neither symmetric nor skew-symmetric, the required probability is =

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$$\frac{3^9-3^6-3^3+1}{3^9}$$

53) $\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\pi-2\tan^{-1}x$ when $x\geqslant 1$

Thus, both statements are true and the second statement is the correct explanation of the first statement.

54) $|z-2-3i|=|(z-5-7i)+(3+4i)|\leqslant|z-5-7i|+|3+4i|$

Thus, $|z-2-3i|\leqslant 9+5=14$

55)

The triangle is isosceles as well as right-angled. In such triangle, the area is equal to the square of length of altitude from the vertex upon the base.

Thus, Area $=\left|\frac{2(0)+3(0)-6}{\sqrt{2^2+3^2}}\right|^2=\frac{36}{13}$ sq. units

56) The differential equation can be re-written as

$$-\left(\frac{ydx-xdy}{y^2}\right)=(xdy+ydx)\sin(xy)$$

or $$-d\left(\frac{x}{y}\right)=\sin(xy).d(xy)$$

Integrating both sides, we get

$$-\frac{x}{y}=-\cos(xy)+c$$

Since, the curve passes through (0,1), therefore putting x = 0 , y = 1 gives c = 1

57) Consider the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

Since, the latus rectum subtends a right angle at the other focus, therefore $\frac{b^2}{a}=2ae$

Thus, $\frac{b^2}{a^2}=2e$ or $e^2-1=2e$. Then, $e^2-1=0$ which gives $e=\sqrt{2}+1$

58) We represent the terms of the progressions through the following table:

Terms	1	2	3	4	5	6	7	8	9
AP	$a-3d$	$a-2d$	$a-d$	a	$a+d$	$a+2d$	$a+3d$	$a+4d$	$a+5d$
GP	a/r^3	a/r^2	a/r	a	ar	ar^2	ar^3	ar^4	ar^5

Given, $a-3d=-1\Rightarrow d=\frac{a+1}{3}$

Also, $ar^4=80\Rightarrow r^2=\sqrt{\frac{80}{a}}$

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And, $(a+2d)+(a+3d)=ar^2 \Rightarrow 2a+5d=ar^2$

Eliminating d and r, we get the equation

$2a+5\frac{(a+1)}{3}=a\sqrt{\frac{80}{a}}$

Squaring the equation, we get

$12la^2-610a+25=0 \text{ or } a=5, a=\frac{5}{121}$

For $a=\frac{5}{121}$, the 9th term is not an integer

For $a=5, r^2=\sqrt{16}$ and $r=\pm 2$ and the smallest 9th term is -160

59) (P) $\sin x+\cos x=1 \Rightarrow \cos\left(x-\frac{\pi}{4}\right)=\frac{1}{2} \Rightarrow x-\frac{\pi}{4}=2nx\pm\frac{\pi}{4}=2n\pi \text{ or } x=2n\pi+\frac{\pi}{2}$

Thus, there are 2 solutions in the given interval

(Q) $\sqrt{3}\tan(3x)=1 \Rightarrow \tan(3x)=\frac{1}{\sqrt{3}} \Rightarrow 3x=n\pi+\frac{\pi}{6} \Rightarrow x=\frac{(6n+1)\pi}{18}$

Thus, there are 2 solutions in the given interval

(R) $2\cos(2x)=\sqrt{3} \Rightarrow \cos 2x=\frac{\sqrt{3}}{2} \Rightarrow 2x=2n\pi\pm\frac{\pi}{6}=x=\frac{(12n\pm 1)\pi}{12}$

Thus, there are 6 solutions in the given interval

(S) $\cos x-\sin x=-1 \Rightarrow \cos\left(x+\frac{\pi}{4}\right)=-\frac{1}{\sqrt{2}} \Rightarrow x+\frac{\pi}{4}=2n\pi\pm\frac{3\pi}{4}$

$\Rightarrow x=2n\pi+\frac{\pi}{2} \text{ or } x=(2n+1)\pi$. Thus, there are 4 solutions in the given interval.

60) $\frac{3}{1+x^2}\in(2,3) \text{ when } x\in\left(0,\frac{1}{\sqrt{2}}\right)$

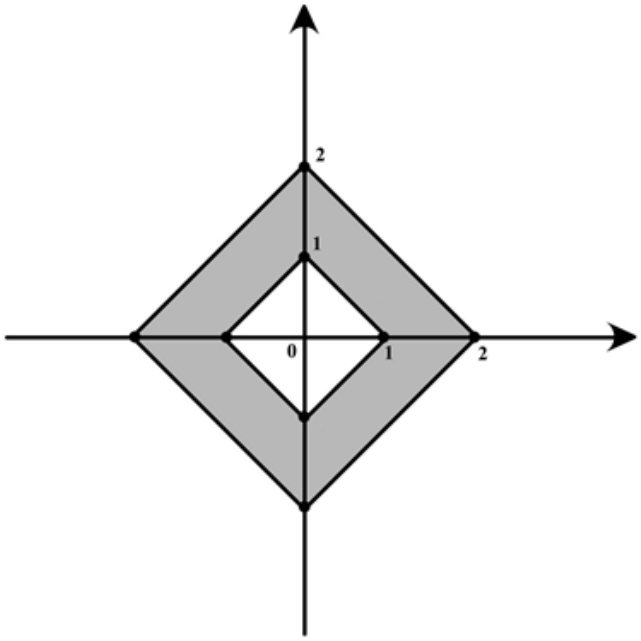
$\frac{3}{1+x^2}\in(1,2) \text{ when } x\in\left(\frac{1}{\sqrt{2}},\sqrt{2}\right)$

$\frac{3}{1+x^2}\in(0,1) \text{ when } x\in\left(\sqrt{2},\infty\right)$

Thus, $\int_0^{\infty}\left[\frac{3}{1+x^2}\right]dx=\int_0^{\frac{1}{\sqrt{2}}}2dx+\int_{\frac{1}{\sqrt{2}}}^{\sqrt{2}}1dx+\int_{\sqrt{2}}^{\infty}0\,dx=\sqrt{2}+\sqrt{2}-\frac{1}{\sqrt{2}}=\frac{3}{\sqrt{2}}$

61) The area of the figure would remain unchanged by shifting of origin. Therefore, replace $x-2\rightarrow X \text{ and } y+1\rightarrow Y$. The new region is now represented by the inequility $1\leqslant|X|+|Y|\leqslant 2$ This area is the area between two squares repesented by $|X|+|Y|=1 \text{ and } |X|+|Y|=2$ which is shaded in the figure below :

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The required area $= 4\left(\frac{1}{2}\times 2\times 2\right) - 4\left(\frac{1}{2}\times 1\times 1\right) = 6\,\text{sq. units}$

62) The equation of family of circles touching the given line at (1,1) is given by
$(x-1)^2+(y-1)^2+\lambda(x+y-2)=0$ or $x^2+y^2+(\lambda-2)x+(\lambda-2)y+(2-2\lambda)=0$
Using condition of orthogonality with the circle $x^2+y^2+2x+2y-2=0$, we get
$(\lambda-2)+(\lambda-2)=(2-2\lambda)+(-2)\Rightarrow\lambda=1$
Therefore, the equation of circle S is $x^2+y^2-x-y=0$

63) $f'(x)=3x^2+2ax+b$
The function f (x) will be strictly increasing when $D\leqslant 0$ i.e. $a^2\leqslant 3b$
The favourable ordered pairs (a,b) are
$(1,1),(2,2),(3,3),(1,4),(2,4),(3,4),(1,5),(2,5),(3,5),(1,6),(2,6),(3,6),(4,6)$ i.e. 16 pairs
The required probability is $\frac{16}{36}$ i.e. $\frac{4}{9}$

64)

The focus of the parabola is (4,0). Focus lies on the director circle of the given circle and hence the slope of this focal chord must be 1. The length of this focal chord will be $16.\text{cose}\text{s}^2 45^\circ$ which is 32 units.

65) Both statements are true only when A is a finite set. Consider the following functions :
$f:R\to R, f(x)=e^x$ is one-one but not onto and $f:R\to R, f(x)=x^3-x$ is onto but not one-one

66)

We know that the sum of focal distances of any point on the ellipse is equal to length of major

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axis (i.e. 2a) and the distance between foci is 2ae.

Thus, eccentricity $e = \frac{2}{2\sqrt{2}+2} = \frac{1}{\sqrt{2}+1} = \sqrt{2}-1$

67)

$$\sum_{i=1}^{10}(x_i - 5) = 10$$

$$x_i - 5 = \frac{1}{10}\sum_{i=1}^{3}(x_i - 5) = 1$$

⇒ Mean of observation

⇒ μ = mean of observation $(x_i - 3)$

= (mean of observation $(x_i - 5)$) + 2

= 1 + 2 = 3

Variance of observation

$$x_i - 5 = \frac{1}{10}\sum_{i=1}^{10}(x_i - 5)^2$$

$-$ (Mean of $(x_i - 5))^2 = 3$

⇒ λ = variance of observation $(x_i - 3)$

= variance of observation $(x_i - 5) = 3$

∴ (μ, λ) = (3, 3)

68) Using the identity : $\tan\theta + \tan(\theta + 60^\circ) + \tan(\theta - 60^\circ) = 3\tan 3\theta$ and putting the value of $\theta = 10^\circ$, we get the value as $3\tan 30^\circ$ or $\sqrt{3}$

69) The system has non-trivial solutions if

$$\begin{vmatrix} p+a & b & c \\ a & q+b & c \\ a & b & r+c \end{vmatrix} = 0 \Rightarrow aqr + bpr + pqr = 0 \Rightarrow \frac{a}{p} + \frac{b}{q} + \frac{c}{r} = -1$$

70) $x^2 + px + q = 0$ has roots r, s and $x^2 + rx + s = 0$ has roots p, q

$r + s = -p$ (1)

$rs = q$ (2)

$p + q = -r$ (3)

$pq = s$ (4)

$(2) \times (4) \Rightarrow pqrs = qs \Rightarrow pr = 1$ (5)

$(1) + (3) \Rightarrow p + q + r + s = -p - r$

$\Rightarrow 2(p + r) + q + s = 0$ (6)

Now p is a root of

$x^2 + rx + s = 0 \Rightarrow p^2 + rp + s = 0$

And r is a root of $x^2 + px + q = 0 \Rightarrow r^2 + pr + q = 0$

Adding those two

$\Rightarrow (p + r)^2 - 2(p + r) = 0 \Rightarrow p + r = 0, p + r = 2$

Also $pr = 1 \Rightarrow p = 1, r = 1$

From (4) $pq = s \Rightarrow q = s$

$\therefore q + s = -2(p + r)$

$\Rightarrow q + s = -4 \Rightarrow q = s = -2$

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$\therefore (4p+q+2r+s) = 4-2+2-2 = 2$

71)

$$\lim_{x\to 2} \frac{2^x + 2^{3-x} - 6}{2^{-\frac{x}{2}} - 2^{1-x}} = \lim_{x\to 2} \frac{2^{2x} - 6.2^x + 8}{2^{\frac{x}{2}} - 2} = \lim_{x\to 2} \frac{(2^x - 4)(2^x - 2)}{2^{\frac{x}{2}} - 2} = \lim_{x\to 2} \left(2^{\frac{x}{2}} + 2\right)(2^x - 2) = 8$$

72) If θ is the angle between unit vectors $\vec{a}$ and $\vec{b}$, then we have

$$\left|\vec{a} - \vec{b}\right| = \sqrt{1 + 1 - 2\cos\theta} = 2\sin\frac{\theta}{2} \text{ and } \left|\vec{a} + \vec{b}\right| = \sqrt{1 + 1 + 2\cos\theta} = 2\cos\frac{\theta}{2}$$

Therefore, $3\left|\vec{a} - \vec{b}\right| + \frac{7}{2}\left|\vec{a} + \vec{b}\right| = 6\sin\frac{\theta}{2} + 7\cos\frac{\theta}{2} \in \left[-\sqrt{85}, \sqrt{85}\right]$

Hence, the maximum possible integral value is 9

73) Shortest distance between lines

$$= \left| \frac{\left(\left(3\hat{i} + 2\hat{k}\right) - \left(\hat{i}\right)\right) \cdot \left(\left(2\hat{i} + 3\hat{j} + 2\hat{k}\right) \times \left(2\hat{i} + 5\hat{j} + 4\hat{k}\right)\right)}{\left|\left(2\hat{i} + 3\hat{j} + 2\hat{k}\right) \times \left(2\hat{i} + 5\hat{j} + 4\hat{k}\right)\right|} \right| = 2$$

74) $5^{99} = 5\left(5^{98}\right) = 5\left(25^{49}\right) = 5(26-1)^{49}$

$$= 5\left(\underbrace{{}^{49}C_0 26^{49} - {}^{49}C_1 26^{48} + \ldots + {}^{49}C_{48}26}_{\text{Multiple of 13}} - {}^{49}C_{49}\right)$$

$= 5$ (multiple of 13)$-5 = 5$ (Multiple of 13)$-13 + 8$

Thus, the remainder is 8

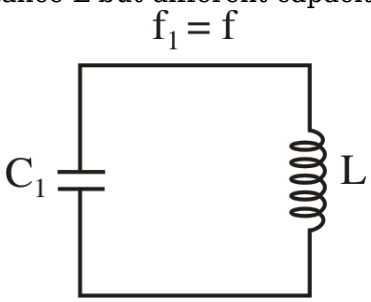
75) Consider both L's to be distinct. The number of such words is then $3 \times 2 \times 3! = 36$. Since, both L's are now idnetical, therefore the number of such words must be $\frac{36}{2!} = 18$

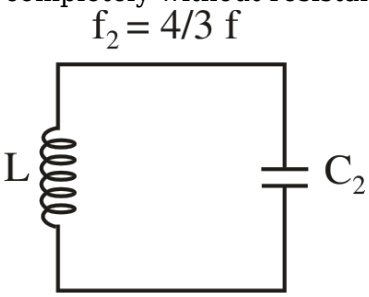
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PART-1 : PHYSICS

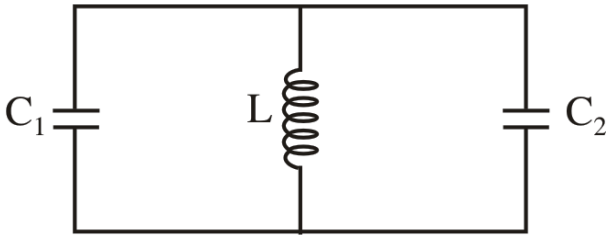
SECTION-I (i)

1) A circuit consisting of an ideal coil and an ideal capacitor is called a resonant circuit. The two electrical resonant circuits shown, with identical inductance L but different capacitances C, oscillate completely without resistance at the given frequencies.





What is the oscillation frequency f_{12} (natural frequency) of the following coupled system?



(A) $\frac{2}{3}f$

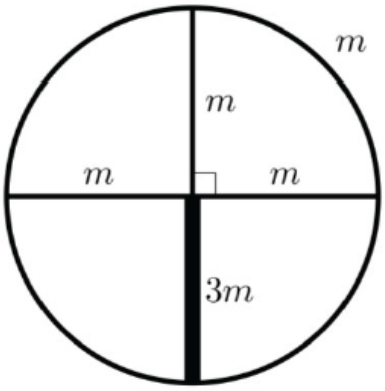
(B) $\frac{3}{4}f$

(C) $\frac{4}{5}f$

(D) $\frac{5}{4}f$

2) A wheel of radius R has a thin rim and four spokes, each of which have uniform density. The entire rim has mass m. three of the spokes each have mass m. and the fourth spoke has mass 3m. The wheel is suspended on a horizontal frictionless axle passing through its center. If the wheel is slightly rotated from its equilibrium position, what is the angular frequency of small oscillations?

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(A) $\sqrt{\frac{g}{3R}}$

(B) $\sqrt{\frac{g}{2R}}$

(C) $\sqrt{\frac{2g}{3R}}$

(D) $\sqrt{\frac{g}{R}}$

3) An empty freight car on a level railroad track has a mass M. A chute above the freight car opens and grain falls down into the car at a rate of r, measured in kilograms per second. The grain falls a vertical distance h before hitting the bed of the freight car without bouncing up. How much normal force is exerted on the freight car from the rails at a time t after the grain begins to hit the bed of the car? Assume the grain starts from rest.

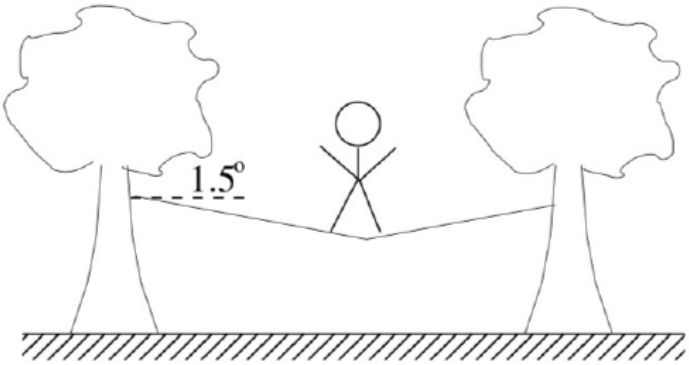
(A) Mg + rtg

(B) $Mg + r\sqrt{gh}$

(C) $Mg + r\sqrt{gh} + rtg$

(D) $Mg + r\sqrt{2gh} + rtg$

4) A massless cable of diameter 2.54 cm (1 inch) is tied horizontally between two trees 18 m apart. A tightrope walker stands at the center of the cable, giving it a tension of 7300 N. The cable stretches and makes an angle of 1.50° with the horizontal. The Young's modulus of the cable is:



(A) 1.5×10^6 N/m^2

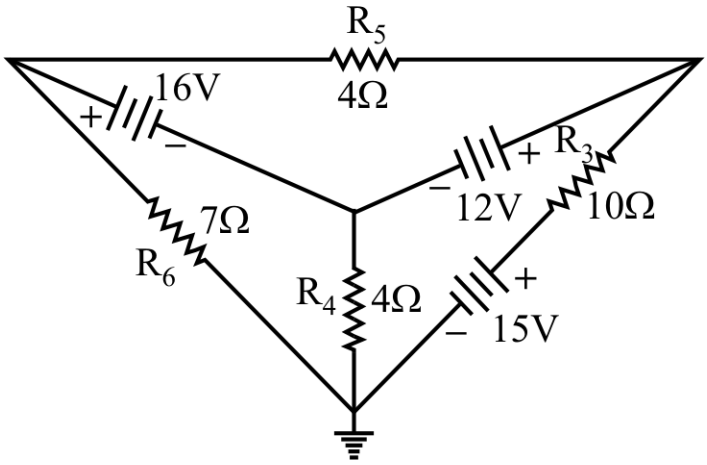
(B) 2.2×10^9 N/m^2

(C) 2.4×10^{10} N/m^2

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(D) 4.2×10^{10} N/m^2

SECTION-I (ii)



1) For the circuit shown in the figure :

(A) Current through R_5 is approximately 4 A.

(B) Current through 12V battery is approximately 3.2 A.

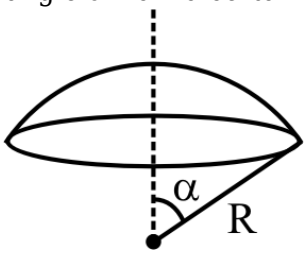
(C) Current through R_6 is approximately 1.7 A

(D) Current through R_3 is approximately 0.7 A.

2) An insulating sphere of radius R is coated with a uniform surface charge density σ on its exterior surface.

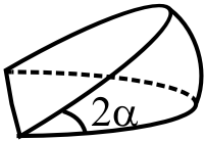
Case-I :-

Suppose that we cut off a spherical cap corresponding to a half-angle α from a certain axis (i.e. the cap has base radius R sinα) and remove the rest of the sphere.



Case-II :-

A spherical wedge of half-angle α (a slice of watermelon) is extracted from the sphere and the rest of the sphere is removed instead.



(A) The electric field at the center of original sphere in case-I is $\frac{\sigma \sin\alpha}{4\epsilon_0}$.

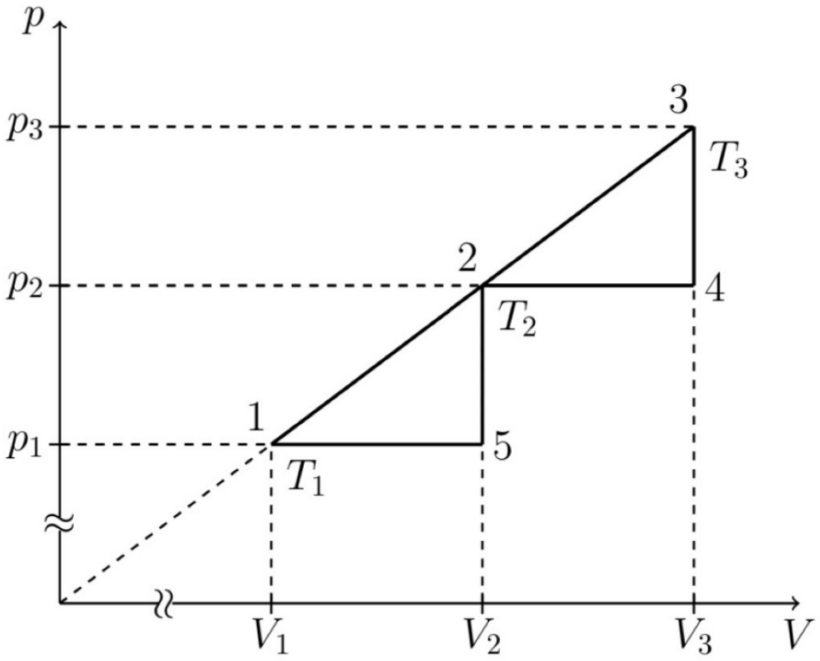
(B) The electric field at the center of original sphere in case–I is $\frac{\sigma \sin^2\alpha}{4\epsilon_0}$.

(C) The electric field at the center of original sphere in case–II is $\frac{\sigma \sin\alpha}{4\epsilon_0}$.

(D) The electric field at the center of original sphere in case II is $\frac{\sigma \sin^2\alpha}{4\epsilon_0}$.

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3) A heat engine, with an ideal gas composed of diatomic molecules as the working substance, operates in a cycle 1 → 2 → 3 → 4 → 2 → 5 → 1, whose pV-diagram is shown in Figure. Points 1, 2, and 3 lie on a straight line passing through the origin, with point 2 being the midpoint of segment 1 → 3. The lowest temperature in the cycle is T_{min}, and the maximum temperature of the cycle is k-times higher. Take T_{min} = 300 K and k = 4.



(A) The value of T_2 is 675 K.

(B) The ratio $\frac{V_3}{V_1}$ is 2.

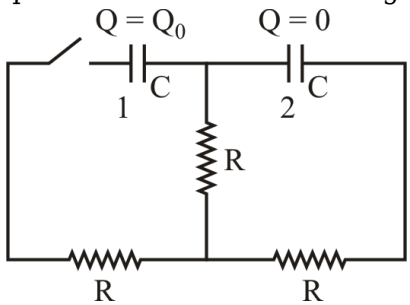
(C) The efficiency of cyclic process is $\frac{25}{9}\%$.

(D) The efficiency of cyclic process is $\frac{25}{18}\%$.

SECTION-II (i)

Common Content for Question No. 1 to 2

The capacitor 1 has an initial charge Q_0 and capacitor 2 is initially uncharged. The switch is closed at t = 0.



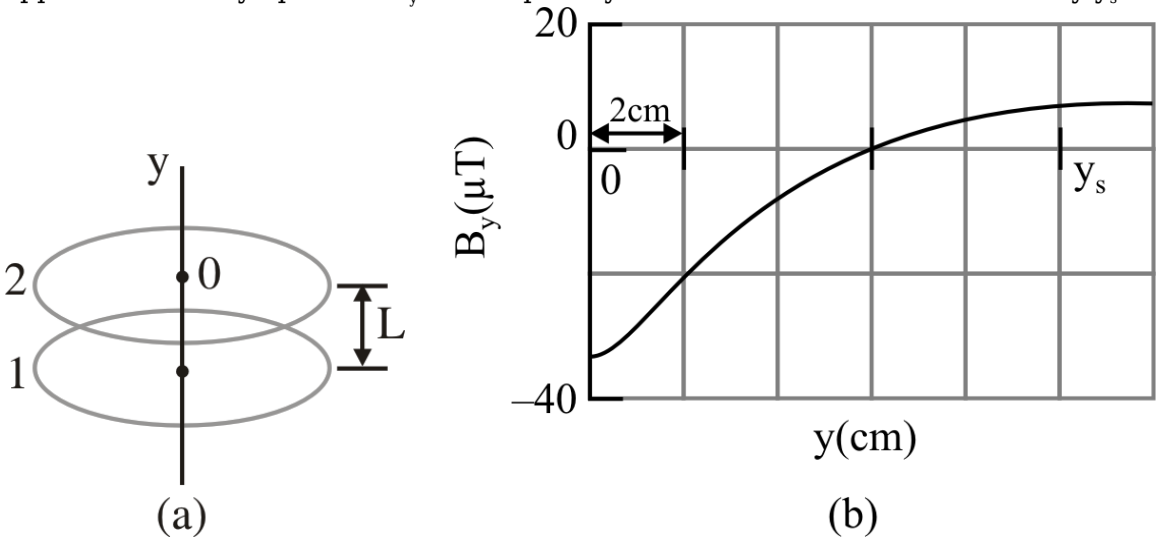
1) The maximum charge on capacitor 2 is $\frac{Q_0}{\sqrt{\alpha}}$. The value of α is.

2) The time at which the charge on capacitor 2 is maximum is αRC□nβ. The value of α + β is

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Common Content for Question No. 3 to 4

In figure-(a), two circular loops, with different currents but the same radius of 4.0 cm, are centered on a y axis. They are initially separated by distance L = 3.0 cm, with loop 2 positioned at the origin of the axis. The currents in the two loops produce a net magnetic field at the origin, with y component B_y. That component is to be measured as loop 2 is gradually moved in the positive direction of the y axis. Figure-(b) gives B_y as a function of the position y of loop 2. The curve approaches an asymptote of B_y = 7.20 μT as y → ∞. The horizontal scale is set by y_s = 10.0 cm.

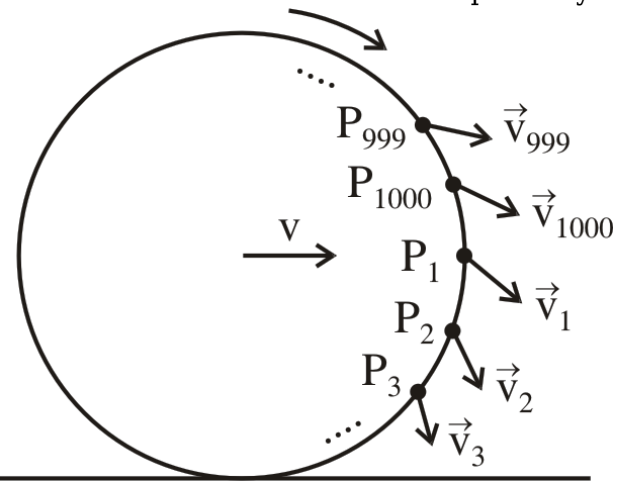


3) The current in loop-2 (in A) is :

4) The current in loop-1 (in A) is :

SECTION-II (ii)

1) A rigid ring rolls without slipping along horizontal ground, with constant translational velocity v = 5.0 m s^{-1} towards the right. Consider 1000 points P_1, P_2, ... , -P_{1000} on the ring, evenly spaced across the ring's full circumference, with the first point P_1 taken to be the ring's rightmost point. Denote their velocities as $\vec{v}_1, \vec{v}_2, \ldots\ldots, \vec{v}_{1000}$ respectively. The value of $|\vec{v}_1 + \vec{v}_2 + \vec{v}_3 \ldots\ldots\ldots \vec{v}_{1000}|$ (in m/s)

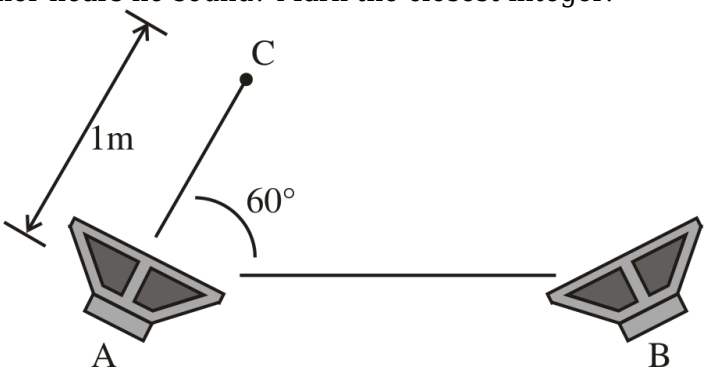


is

2) The drawing shows a loudspeaker A and point C, where a listener is positioned. A second loudspeaker B is located some-where to the right of loudspeaker A. Both speakers vibrate in phase and are playing a 68.6-Hz tone. The speed of sound is 343 m/s. What is the closest (in m) to speaker

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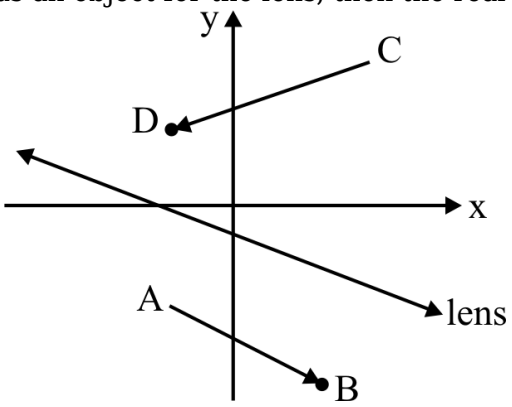
A that speaker B can be located, so that the listener hears no sound? Mark the closest integer.



(Assume both speakers behave as point sources)

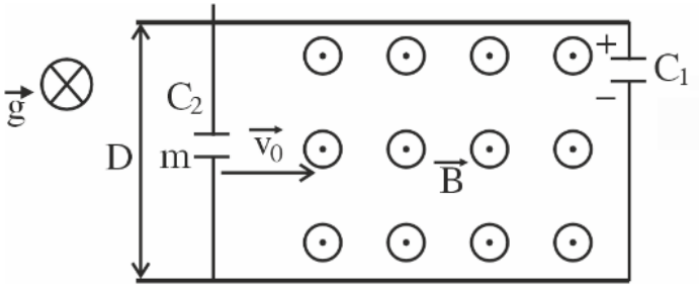
3) Three perfect linear polarizers are stacked normal to a central axis along which is incident a beam of natural light of intensity I_0. If the first and last polarizers are crossed and if the middle one rotates at a rate 2π rad/s about the axis. Find the frequency (in 1/sec) of oscillation of intensity of emerging light?

4) A convergent ideal thin lens is placed such that the plane of the lens is perpendicular to the plane of the paper, and the optical axis lies within the plane of the paper. Define points A(–4.0, –4.2), B(3.0, –9.0), C(10.0, 10.5) and D(–1.5, 4.5) within the plane of the paper. All the coordinates given are in cm. If the vector $\overrightarrow{AB}$ is used as an object for the lens, then the real image $\overrightarrow{CD}$ is formed. Find



the focal length of the lens (in cm).

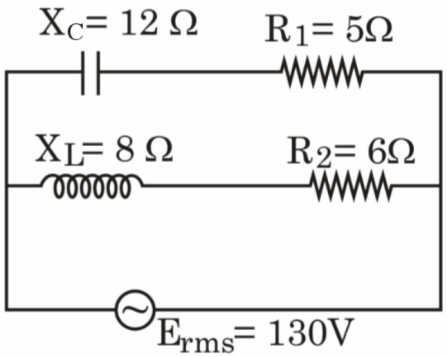
5) Two sufficiently long horizontal rails with zero resistance are connected to a charged capacitor with capacitance C_1. The distance between the rails is D. The rails are placed in a vertical constant magnetic field with an induction of B. Initially, the rails are connected to an uncharged capacitor with capacitance C_2, and this capacitor is simultaneously given an initial velocity v_0 along the rails. The terminals of the capacitor C_2 can slide along the rails without friction or loss of contact. The mass of the capacitor C_2 and its terminals is m. Find the initial charge Q (in μC) the capacitor C_1, assuming that capacitor C_2 comes to rest during its motion. (Take m = 1kg, v_0 = 2 m/s, D = 1m, C_1 =



1μF, C_2 = 2 μF and B = 1T.

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6) The amount of power delivered by the ac source in the resistance R_1 of the circuit given below is X watt, then X is :-



PART-2 : CHEMISTRY

SECTION-I (i)

1) When diazoaminobenzene is treated with a small quantity of aniline hydrochloride in the presence of anilene, then

(A) Deep-Red coloured product is formed

(B) Azobenzene is formed

(C) Sodium fusion extract of dye formed in the above reaction, fails to produce prussian blue/ green colour with $FeSO_{4(aq)}$ and Conc. H_2SO_4

(D) Orange dye is formed, when product formed in the above reaction is treated with
(i) $NaNO_2$ + HC□, 0°C (ii) H_2O, Δ

2) When equal volumes of two pure liquids phenol and aniline are mixed together, then

(A) Resultant Free energy of the liquid mixture is greater than its free energy in pure states

(B) Solution shows negative deviation from Raoult's law

(C) For this process, 'ΔS' of system is less than zero due to Exothermic nature of this process.

(D) For this process, $\Delta S_{surrounding}$ is less than zero due to exothermic nature of this process.

3) Which one of the following statement regarding potash alum is **INCORRECT** ?

(A) During its preparation, dilute H_2SO_4 is used to prevent the hydrolysis of mixture

(B) Aqueous solution of potash alum gives white precipitate with perchlorate aqueous solution

(C) It is used as Fixing agent in dye industry

(D) This salt shrinks on heating initially.

4) A H-like species emitted a photon corresponding to the first line of Lyman series. The photon liberated a photoelectron from He^+ in ground state. The de-Broglie wavelength of the photoelectron is 2 Å. Then, the atomic number of H-like species is
Given : Energy of electron in ground state of H-atom = – 13.6 eV/atom
h = 6.626 × 10^{-34} J-sec, mass of electron = 9.1 × 10^{-31} kg

(A) 3

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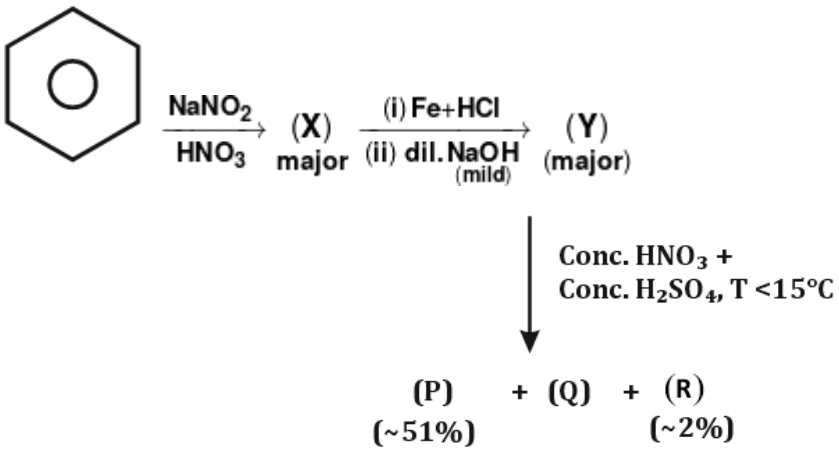
(B) 2

(C) 4

(D) 6

SECTION-I (ii)

1)



(P), (Q) and (R) are isomers

Which of the following statement(s) is / are **CORRECT** ?

(A) Boiling point order : (P) > (Q) > (R)

(B) Basicity order : (R) > (Q) > (P)

(C) Melting point order : (P) > (Q) > (R)

(D) 'X' gives grey precipitate or silver mirror, when it reacts with (i) $Zn/NH_4C\square$ and (ii) $[Ag(NH_3)_2]$ OH, Δ

2) At 25°C, the dissociation constant (K_a) of weak mono basic acid (HA) is numerically equal to the dissociation constant (K_b) of its conjugate base (A^-). Then, identify the **CORRECT** option(s)

(A) $K_a(HA) = 10^{-7}$

(B) pH of 0.1M HA aqueous solution ≈ 3.0

(C) pH of 0.1 M aqueous solution of A^- ≈ 10.0

(D) pH of aqueous solution containing 0.1 M HA and 0.01 M HC□ ≈ 2.0

3) A complex salt is represented as $CoC\square_3 . xNH_3$ and it's 0.1 molal solution in water has 'ΔT_f' of 0.372°C.

Given :

Cryoscopic and Ebullioscopic constants for H_2O are 1.86 K-kg mol^{-1} and 0.52 K-kg mol^{-1} respectively.

Assume complex salt undergoes complete ionisation in H_2O

Which of the following statement(s) is/are **CORRECT** regarding this complex salt ?

(A) IUPAC name of this salt is Pentaamminechloridocobalt(III) chloride

(B) Diamagnetic in nature

(C) Total stereomers possible for this salt = 2

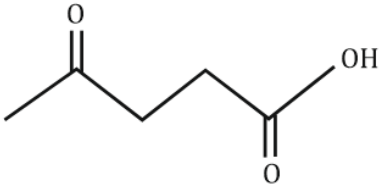
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(D) Given solution will boil between 100°C to 101°C

SECTION-II (i)

Common Content for Question No. 1 to 2

Levulinic acid is a versatile chemical that can be used to make pharmaceuticals, plastics and fuels. It



(Levulinic acid)

can be made from fructose. Furanose form of Fructose can be converted to levulimic acid as follows :

Furanose form of D(–) - Fructose $\xrightarrow{-H_2O}$ (P) (contains aldehyde) + other tri substituted alkenes

(P) $\xrightarrow{-2H_2O}$ (Q) (major) $\xrightarrow{\text{Finally}}$ Levulinic acid + (R)

Given :

'Q' has diatropic nature

(R) gives '+' ve $NaHCO_3$ test and Tollen's test

Atomic mass : C–12, H–1, O–16, Ag–108, N–14

1) (Q) + α-naphthol $\xrightarrow[H_2SO_4]{\text{Conc}}$ Coloured product (S)

Total number of hetero atoms present in one Cationic unit of 'S' is .....

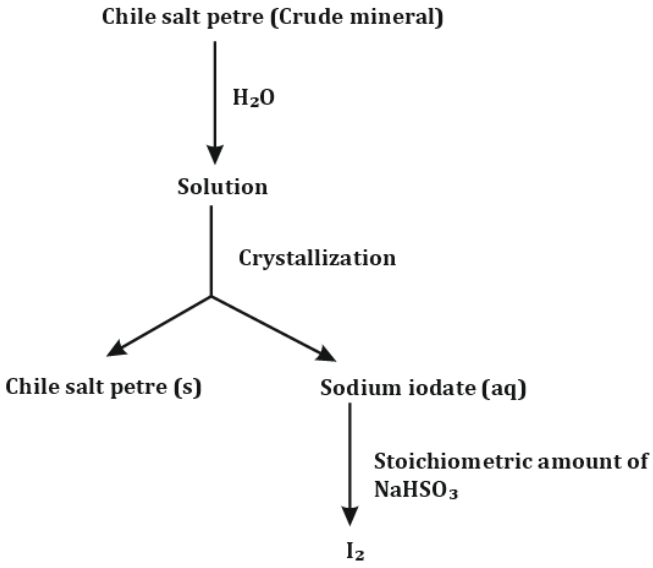
2) (R) $\xrightarrow[\Delta]{[Ag(NH_3)_2]OH}$ Coloured product (X) + other products

The maximum amount of (R) in gram required to consume completely one mole of $[Ag(NH_3)_2]OH$ in the formation of 'X' is .....

Common Content for Question No. 3 to 4

Caliche (Sodium iodate) is the main source for iodine, which is found in nature with chile salt petre

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as crude chile salt petre **Reactions involved in the above process :**

$$IO_3^- + HSO_3^- \rightarrow I^- + SO_4^{2-} + H^+$$
$$I^- + IO_3^- + H^+ \rightarrow I_2 + H_2O$$

[Given : Atomic mass : Na = 23, S = 32, O = 16, H =1, I = 127]

3) Determine equivalent mass of I_2 (in gm $eq.^{-1}$) produced when iodide salt is treated with iodate salt in acidic medium

4) One litre of aqueous solution contains 5 gram of $NaIO_3$ is treated with stoichiometric quantity of $NaHSO_3$. Now an additional amount of same solution ($NaIO_{3(aq)}$) is added to resultant mixture to convert iodide into Iodine. What volume of solution **(in mL)** containing $NaIO_3$ must be added in step-II to bring in complete conversion of I^- to I_2

SECTION-II (ii)

1) (i) $PCl_3 + LiH \rightarrow (X)\uparrow + Y$

(ii) $(X) \xrightarrow[H_2SO_4]{HCHO\ (excess)} Salt\ \ (Z)$

white solid which is used for making fire-proof cotton fabrics

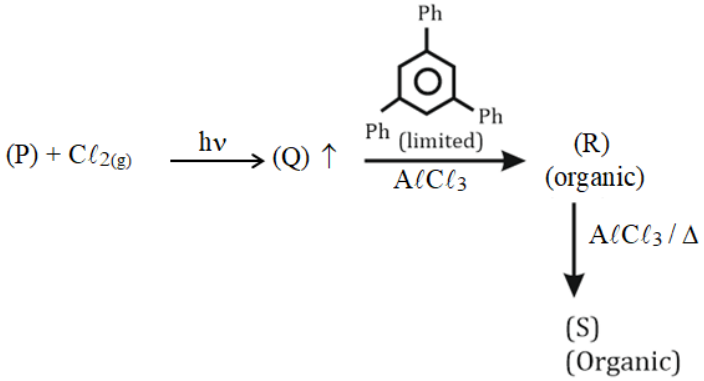
The maximum amount (in gram) of **(Z)** formed when 13.75 gr of PC□$_3$ is used in the above reaction sequence is **(Nearest Integer)**

Given : % yield of reaction (i), (ii) are 100 and 50 respectively

Atomic mass : P-31, C□-35.5, Li-7, H-1 , C-12, O-16, S-32

2) $K_4[Fe(CN)_6] + \underset{(conc)}{H_2SO_4} + H_2O \rightarrow (P)\uparrow + \text{other products}$

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Then, D.U of product (S) is

Given : (P) and (Q) are poisonous gases

Consider major products in all above reactions

3)

$K_2Cr_2O_7$ is a good oxidising agent in acidic medium. By how much **(in volt)** oxidising power of $K_2Cr_2O_7$ decreases when pH changes from 0 to 14 at 298 K. Assume that activities of all other components are constant at unity. **(If the answer is X volt then find the value of 100X)**

Given : $\frac{2.303RT}{F} = 0.06\,\text{Volt}$

4) One litre of 1M H_2O_2 solution was left open, causing first order decomposition of H_2O_2 with rate constant equal to 6.93×10^{-2} sec^{-1}. Determine amount of energy released **(in kJ)**, if this solution is reacted with excess of $I^-_{(aq)}$ in acidic medium after 20 sec from the start.

Given : Assume no change in volume of H_2O_2 solution

$\Delta H_f^0\left[H_2O_{2(aq)}\right] = -200\,\text{kJ/mole}$

$\Delta H_f^0\left[I^-_{(aq)}\right] = -60\,\text{kJ/mole}$

$\Delta H_f^0\left[I_{2(aq)}\right] = 60\text{kJ/mole}$

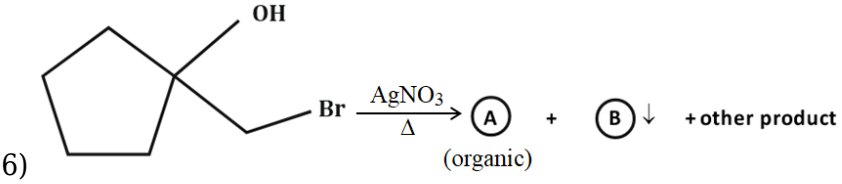
$\Delta H_f^0\left[H_2O_{(\ell)}\right] = -290\,\text{kJ/mole}$

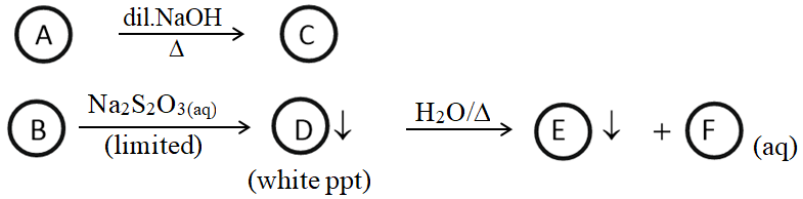
$\Delta H_f^0\left[H^+_{(aq)}\right] = 0$

5) Among the following the total number of species which is / are paramagnetic

$C_2, O_2, S_2, O_2^+, N_2, N_2^-, CN^-, B_2, NO_2, NO^+, C\ell O_2, O_3$

6)





If D.U of product (C) = 'X'

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Spin only magnetic moment of product (E) = 'Y' B.M

Then, the value of [100 Y + X] is

PART-3 : MATHEMATICS

SECTION-I (i)

1) Let $f(x) = x^2 - 3x - 7$. Let g(x) and h(x) be two quadratic polynomials also, both with the coefficient of x^2 equal to 1. Rohan computes each of the three sums f(x) + g(x) , g(x) + h(x) , h(x) + f(x) , and is surprised to find that each pair of these sums has a common root, and these three common roots are distinct. If g(0) = 2, then h(0) is

(A) $\frac{33}{52}$

(B) $\frac{19}{33}$

(C) $\frac{52}{19}$

(D) $\frac{71}{19}$

2) The volume enclosed by the planes $|5x-6| + |10y-7| + |z+4| = 10$ is

(A) $\frac{10}{3}$

(B) $\frac{40}{3}$

(C) $\frac{80}{3}$

(D) $\frac{20}{3}$

3) The area of the region defined as and $y \le e^{||x| \ln |x||}$, $x^2 + y^2 - 2(|x| + |y|) + 1 \ge 0$ and $y \ge 0$, where $|x| \le 1$, if α is the x-coordinate of the point of intersection of curves $y = e^{||x| \ln |x||}$ & $x^2 + y^2 - 2(|x| + |y|) + 1 = 0$ in 1st quadrant, is

(A) $4\left[-\int_0^{\alpha} e^x \ln x\,dx + \int_{\alpha}^{1}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$

(B) $4\left[\int_0^{\alpha} e^x \ln x\,dx - \int_1^{\alpha}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$

(C) $2\left[-\int_0^{\alpha} e^x \ln x\,dx + \int_{\alpha}^{1}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$

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(D) $2\left[\int_0^{\alpha} ex\ln x\ dx + \int_{\alpha}^{1}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$

4) If $a + b + c = 35$, then maximum value of $\sqrt{3a+13} + \sqrt{3b+13} + \sqrt{3c+13}$ is (where $a, b, c \in R^+$)

(A) 12

(B) $12\sqrt{3}$

(C) 6

(D) $6\sqrt{3}$

SECTION-I (ii)

1) Let $A = \begin{bmatrix} \alpha & \beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \alpha+\beta \end{bmatrix}$ and $|2A^3| = 2^{12}$ where $\alpha,\beta \in I$, then possible values of tr(A) are-

(A) 2

(B) –2

(C) 6

(D) –4

2) Let f(x) be a function defined by $f(x) = \int_0^{3\pi/4} \left|x\cos t - 3\sin t + \sqrt{2}x\right| \left(\frac{1+\sqrt{2}\cos t}{\left(\cos t+\sqrt{2}\right)^3}\right) dt$ where x is any real number, then which of the following options is/are true ?

(A) f(x) is continuous and differentiable everywhere

(B) f(x) is continuous for all x but non-differentiable at two values of x

(C) Minimum value of f(x) is $\frac{3}{2}$ for $x \in R$

(D) Minimum value of f(x) is $\frac{3}{4}$ for $x \in R$

3) Let function y = f(x) satisfies the differential equation $x^2\frac{dy}{dx} = y^2e^{\frac{1}{x}}\ (x \neq 0)$ and $\lim_{x\to 0^-} f(x) = 1$. Identify the correct statement(s) :

(A) Range of f(x) is $(0,1) - \left\{\frac{1}{2}\right\}$

(B) f(x) is bounded

(C) $\lim_{x\to 0^+} f(x) = 1$

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(D) $\int_0^e f(x)\,dx > \int_0^1 f(x)\,dx$

SECTION-II (i)

Common Content for Question No. 1 to 2

N dice are rolled once, then the probability of getting sum L is expressed as P_r (L, N), where dice is a r sided dice with faces marked 1, 2, 3, 4, ... r.

1) Value of $9.\sqrt{P_9(15,2)}$ is

2) Value of P_4 (9, 3) – P_4 (9, 4) is

Common Content for Question No. 3 to 4

$$A.\,adj\left(A^2\right) = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

If A is 3 × 3 matrix with real entries such that

If B is inverse of matrix A such that $A = \lambda B^2 + \mu B + \gamma I$ where λ, μ, γ are scalars and I is identity matrix of order 3 × 3

3) The value of μ is :

4) The value of det(adj (adj A)) is :

SECTION-II (ii)

1) In ΔABC, let B(3,3,–9) and AB = AC and altitude AD is $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{3}$. If co-ordinates of C are (α,β,γ), then |α| + |β| + |γ| is

2) Let ellipse $\frac{x^2}{25} + \frac{y^2}{p^2} = 1$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{q^2} = 1$ are confocal and meet at points A,B,C and D. If foci of ellipse subtend 90° at point A and e_1 and e_2 are eccentricities of ellipse and hyperbola then $\left(\frac{1}{e_1^2} + \frac{1}{e_2^2}\right)$ is

3) Consider the following lines :

$L_1 : x - \sqrt{3}y + 3 = 0,\ L_2 : x - 2y + 4 = 0$
$L_3 : x - 5y + 25 = 0,\ L_4 : x + y + 1 = 0$
$L_5 : x - y + 1 = 0,\ L_6 : x - 4y + 16 = 0$

Let H(L_m, L_n, L_k) denotes orthocenter of triangle formed by lines L_m, L_n, L_k and S(L_m, L_n, L_k) denotes circumcircle of triangle formed by lines L_m, L_n, L_k

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If $H(L_1, L_3, L_5) \equiv (x_1, y_1)$ and $(x_3, 0) \in S(L_1, L_3, L_5) \cap S(L_2, L_4, L_6)$, where $x_3 > 0$. Then $x_1^2 + x_3^2$ is

4) Let $n = 2^{31} \cdot 3^{19}$, then number of factors of n^2 that are less than n and are not factor of n, are

5) Let the mean and standard deviation (non-zero) of marks of class A of 100 students be respectively 40 and α, and the mean and standard deviation (non-zero) of marks of class B of n students be respectively 55 and 30 – α. If the mean and variance of the marks of the combined class of 100 + n students are respectively 50 and 350, then the sum of variances of classes A and B is

6) If $\left[2\left(\vec{a} \times \vec{b}\right) \;\; 2\left(\vec{b} \times \vec{c}\right) \;\; 2\left(\vec{c} \times \vec{a}\right)\right] = 800$ and $\vec{b}.\left(\vec{a} \times \left(\vec{b} \times \left(\vec{c} \times \vec{b}\right)\right)\right) = -250$, then $\left|\vec{b}\right|$ is equal to $\left(\left[\vec{a} \;\; \vec{b} \;\; \vec{c}\right] > 0\right)$

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ANSWER KEYS

PART-1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	C	A	D	D

SECTION-I (ii)

Q.	5	6	7
A.	C,D	B,C	A,B,C

SECTION-II (i)

Q.	8	9	10	11
A.	27.00	4.50	2.60 to 2.70	0.86 to 0.90

SECTION-II (ii)

Q.	12	13	14	15	16	17
A.	5000	4	4	3	3	500

PART-2 : CHEMISTRY

SECTION-I (i)

Q.	18	19	20	21
A.	D	B	D	A

SECTION-I (ii)

Q.	22	23	24
A.	A,C,D	A,C,D	B,C,D

SECTION-II (i)

Q.	25	26	27	28
A.	4.00	23.00	152.40	200

SECTION-II (ii)

Q.	29	30	31	32	33	34
A.	10	18	196	50	7	4

PART-3 : MATHEMATICS

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SECTION-I (i)

Q.	35	36	37	38
A.	C	C	C	B

SECTION-I (ii)

Q.	39	40	41
A.	B,C	A,D	A,B,D

SECTION-II (i)

Q.	42	43	44	45
A.	2.00	0.00	3.00	1.00

SECTION-II (ii)

Q.	46	47	48	49	50	51
A.	9	2	2	589	500	5

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SOLUTIONS

PART-1 : PHYSICS

3) The (upward) normal force on the freight car must equal the total downward forces experienced by the freight ear. The freight car experiences three1 downward forces:
The freight car's own weight:
$F_{g,car} = Mg$
The weight of the grain accumulating in the freight car. The grain accumulates at a (mass) rate r, so after a time t the car contains a total grain mass rt. Thus the total weight of the grain after time t is
$F_{g,grain} + rtg$
The force from absorbing the momentum of the falling grain when the grain lands on the cart. Over some time interval Δt, an amount of grain with mass $\Delta m = r\Delta t$ falls onto the cart. The grain falls from a height h, so it lands on the cart with speed $v = \sqrt{2gh}$. Thus the grain carries a momentum
$\Delta p = \Delta m v = r\Delta t\sqrt{2gh}$
Thus the rate at which the momentum is delivered to the cart, or the force experienced by the cart, is

$$F_{collision} = \frac{\Delta p}{\Delta t} = \frac{r\Delta t\sqrt{2gh}}{\Delta t} = r\sqrt{2gh}$$

Thus the normal force, which equals the total downward force on the cart, is
$F_{g,car} + F_{g,grain} + F_{collision} = Mg + rtg + r\sqrt{2gh}$

4) The strain of the cable is

$$\frac{\Delta L}{L} = \frac{\frac{L_0}{\cos\theta} - L_0}{L_0} = \frac{1}{\cos\theta} - 1 \approx \frac{\theta^2}{2}$$

where L_0 is the initial length. Then the Young's modulus is

$$Y = \frac{(7300\ \text{N})}{\pi(0.0127\ \text{m})^2}\left(\frac{\left(\frac{1.50\pi}{180}\right)^2}{2}\right)^{-1} = 4.2 \times 10^{10}\ \text{N/m}^2$$

10) $$B_y = \frac{\mu_0 i_1 R^2}{2\pi(R^2 + z_1^2)^{3/2}} - \frac{\mu_0 i_2 R^2}{2\pi(R^2 + z_2^2)^{3/2}},$$

where $z_1^2 = L^2$ and $z_2^2 = y^2$ (because the central axis here is denoted y instead of z). The fact that there is a minus sign between the two terms, above, is due to the observation that the datum corresponding to By = 0 would be impossible without it (physically, this means that one of the currents is clockwise and the other is counterclockwise).
(a) As $y \to \infty$, only the first term contributes and (with $B_y = 7.2 \times 10^{-6}$ T given in this case) we can solve for i_1. We obtain $i_1 = (45/16\pi)$ A ≈ 0.90 A.
(b) With loop 2 at y = 0.06 m we are able to determine i_2 from

$$\frac{\mu_0 i_1 R^2}{2(R^2 + L^2)^{3/2}} = \frac{\mu_0 i_2 R^2}{2(R^2 + z_2^2)^{3/2}}$$

We obtain

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$$i_2 = \left(\frac{117\sqrt{13}}{50\pi}\right) A \approx 2.7A$$

14)

Intensity of light after passing through first polarizer, $I_0 = \frac{I_0}{2}$ intensity of light after passing through second polarizer, $I_2 = \frac{I_0}{2}\cos^2\omega t$ intensity of light after passing through third polarizer, $I_3 = \frac{I_0}{2}\cos^2\omega t.\sin^2\omega t$

$I_3 = \frac{I_0}{16}(1-\cos 4\omega t)$

$2\pi f = 4\times 2\pi \Rightarrow f = 4/\text{sec}$

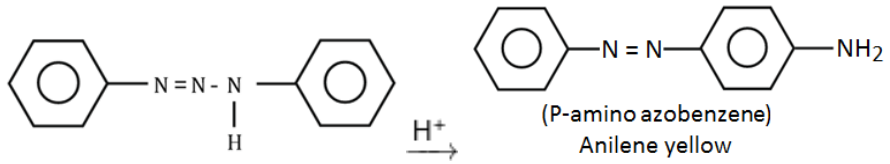
17) Power = $V_{rma} \times I_{rm} \cos\phi$

$= 130 \times \frac{130}{13}.\frac{5}{13} = 500\text{watt}$

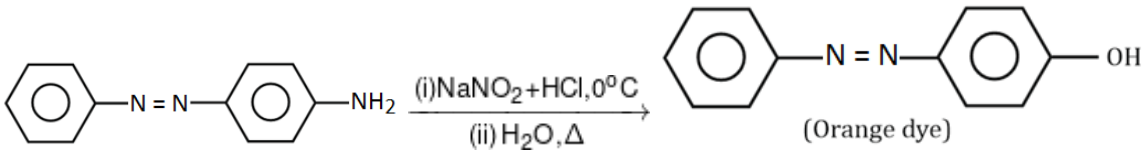
PART-2 : CHEMISTRY

18)

In mild acidic condition, N–N coupling azo compound rearranges to C – N coupling compound.



(P-amino azobenzene)
Anilene yellow



(Orange dye)

19)

(Phenol + Anilene) shows –ve deviation

$\therefore \Delta H_{mix} < 0$

$\Delta S_{sys} > 0$

$\Delta S_{sur} > 0$

$\Delta G_{sys} < 0$

20)

A. Potash alum - $K_2SO_4 . A\square_2(SO_4)_3 . 24H_2O$

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B. Potash $alum_{(aq)} \xrightarrow{C\ell O_{4(aq)}^{-}} \underset{\text{Whiteppt}}{KC\ell O_4} \downarrow$

C. Alum forms $A\square(OH)_3$ on the cloth and dye sticks to this $A\square(OH)_3$

D. Alum swells on heating due to loss of water

21) $\lambda = \sqrt{\frac{150}{V}}\text{Å} \Rightarrow V = \frac{150}{4} = 37.5$

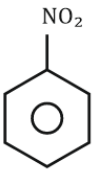
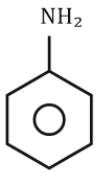
K.E of photo electron = 37.5 eV

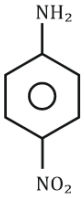
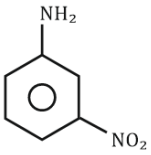
Energy required to remove electron $= \frac{13.6}{1} \times 4 = 54.4$ eV

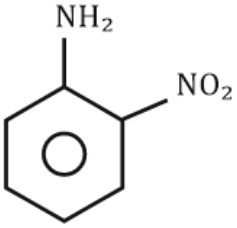
Total energy of photon = 91.9 eV

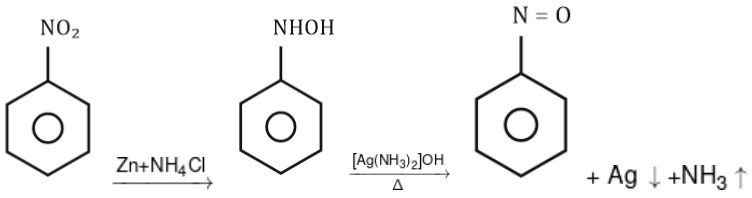
Now, $10.2\ Z^2 = 91.9$

$\Rightarrow Z = 3$

22) 'X' =  ; 'Y' =  ;

'P' =  ; 'Q' =  ;

'R' = 



23) (A) $K_a \cdot K_b = K_w \Rightarrow K_a = K_b = 10^{-7}$ (B) $[H^+] = \sqrt{K_a.C} = \sqrt{10^{-8}} = 10^{-4}M$

pH = 4

(C) $A^- + H_2O \rightleftharpoons HA + OH^-$

$[OH^-] = \sqrt{K_b.C} = \sqrt{10^{-8}} = 10^{-4}M$

pOH = 4 ⇒ pH = 10.0

(D) $[H^+]_{Net} = [H^+]_{HA} + [H^+]_{HCl}$

$\approx [H^+]_{HC\ell} = 0.01M$

pH = 2.0

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24) $\Delta T_f = i.\ K_f\ .$ molality

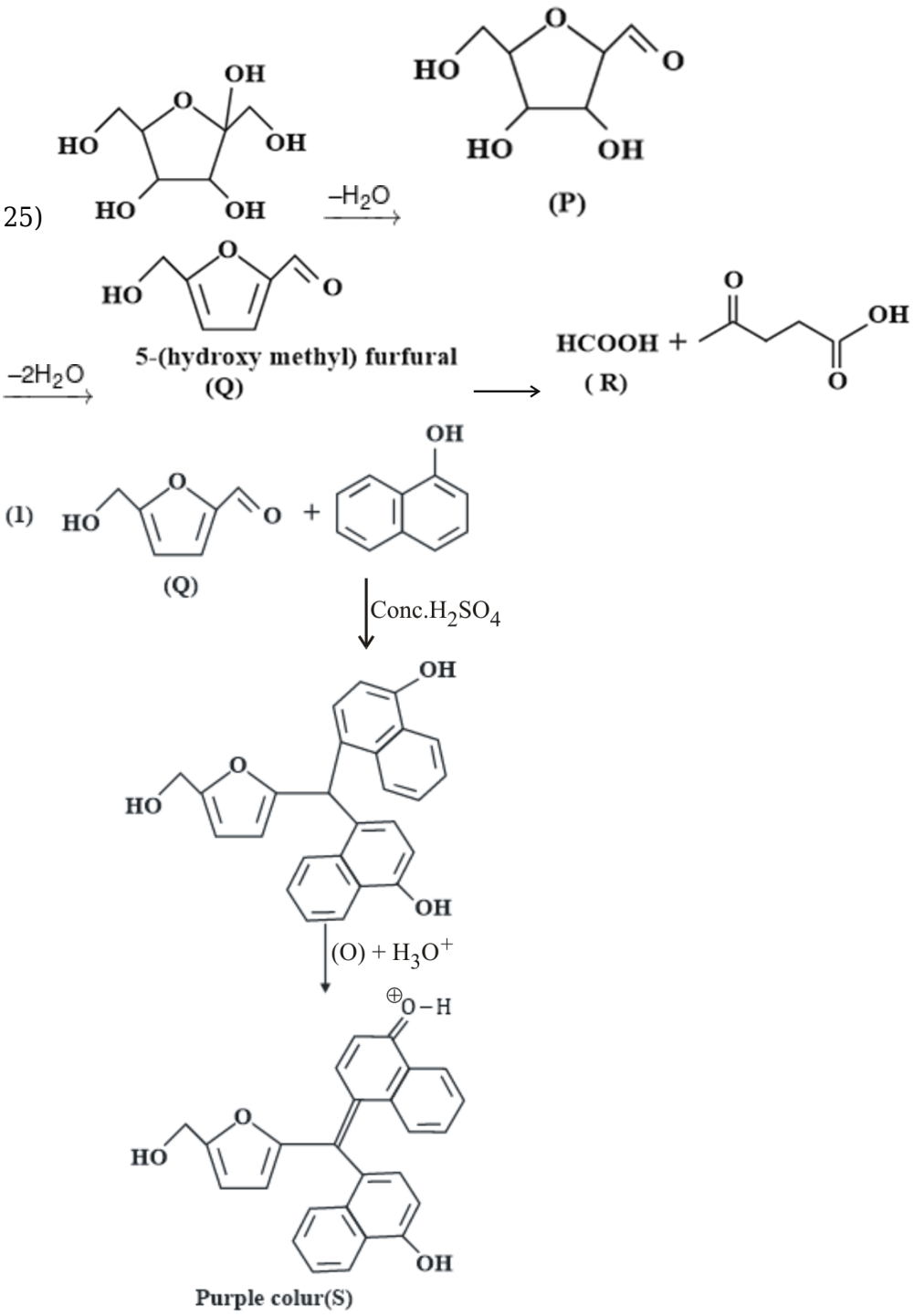
$0.372 = i \times 1.86 \times 0.1$

$i = 2$

Complex salt is $[Co(NH_3)_4C\square_2]\ C\square$

This salt has two G.I, but both isomers are optically inactive.

25)



26) $HCOOH \xrightarrow[\Delta]{Ag(NH_3)_2OH} \underset{(X)}{Ag\downarrow} + NH_3\uparrow + CO_2\uparrow + H_2O$

$H-\overset{\overset{O}{\|}}{C}-OH \rightarrow CO_2 + 2e^-$

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$Ag^+ + e^- \rightarrow Ag$

0.5 mole of HCOOH can reduce one mole of $[Ag(NH_3)_2]OH$

$W_{HCOOH} = 0.5 \times 46 = 23$

27) $5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

n-factor of H_2O = 5/3 = 1.67

28) (i) $IO_3^- + 3HSO_3^- \rightarrow I^- + 3SO_4^{2-} + 3H^+$

moles of $NaIO_3$ taken $= \frac{5}{M_{NaIO_3}}$ = moles of NaI formed

(ii) $5I^- + IO_3^- + 6H^+ \rightarrow I_2 + 3H_2O$

moles of $NaIO_3$ required

$= \frac{5}{5 \times M_{NaIO_3}} = \frac{1}{M_{NaIO_3}}$

Weight of $NaIO_3$ = 1

Volume of $NaIO_3$ required = 1000/5 = 200 mL

29) (i) $PC\ell_3 + 3LiH \rightarrow \underset{(P)}{PH_3} \uparrow + \underset{(Q)}{3LiC\ell}$

(ii) $PH_3 + \underset{(excess)}{HCHO} \xrightarrow{H_2SO_4} \underset{(R)}{[P(CH_2OH)_4]_2 SO_4}$

n_{PCl_3} taken $= \frac{13.75}{137.5} = 0.1$

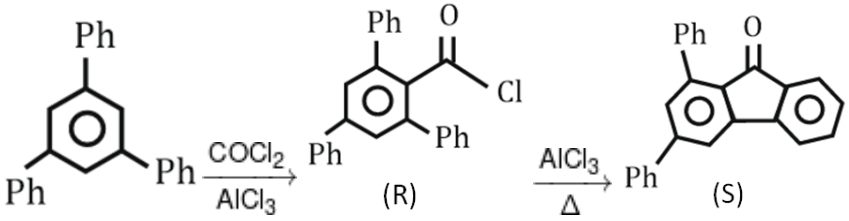
n_{PH_3} formed = 0.1

n_{salt} formed $= \frac{0.1}{2} \times \frac{50}{100} = 0.025$

$W_{salt} = 0.025 \times 406 = 10.15$ g

30) $K_4[Fe(CN)_6] + H_2SO_{4(Conc)} + H_2O \rightarrow K_2SO_4 + FeSO_4 + (NH_4)_2SO_4 + \underset{(P)}{CO} \uparrow$

$CO_{(g)} + C\ell_{2(g)} \xrightarrow{h\nu} \underset{(Q)}{COC\ell_{2(g)}}$



31) $Cr_2O_{7(aq)}^{2-} + 14H_{(aq)}^+ + 6e^- \rightarrow 2Cr_{(aq)}^{3+} + 7H_2O_{(\ell)}$

$$E = E^o - \frac{0.06}{6} \log \frac{[Cr^{3+}]^2}{[Cr_2O_7^{2-}][H^+]^{14}}$$

$$E = E^o + 14(0.01) \log [H^+]$$

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On changing pH from 0 to 14, $[H^+]$ changes from 1M to 10^{-14}M

$E_1 = E^oV$ and

$$E_2 = E^o - 14 \times 0.01 \times 14 = \left(E^0 - 1.96\right) V$$

$\therefore \Delta E = 1.96$ Volt

32) $2H_2O_2 \rightarrow 2H_2O + O_2$

Initial no.of moles of $H_2O_2 = 1$

$$t_{\frac{1}{2}} = \frac{0.693}{k} = 10 \text{ sec}$$

No.of moles of H_2O_2 after 20 sec = 1/4

$H_2O_2 + 2H^+ + 2I^- \rightarrow I_2 + 2H_2O$

$\Delta H^o_r = [60 + 2(-290)] - [-200 + 2(0) + 2(-60)]$

= – 520 + 320

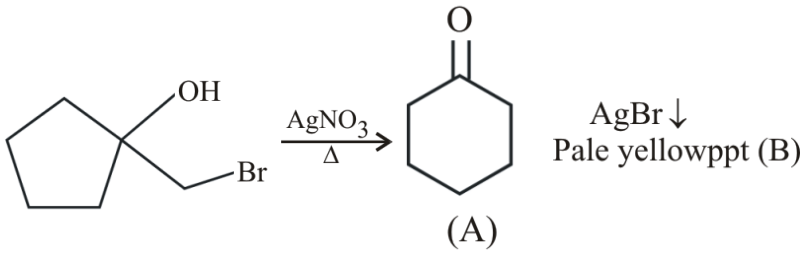
= – 200 kJ

Amount of heat released $= 200 \times \frac{1}{4} = 50\,kJ$

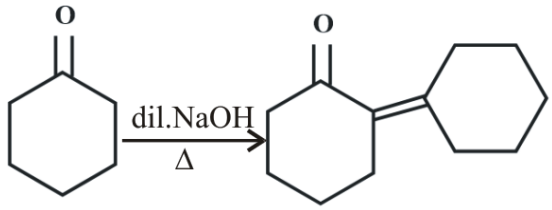
33) The following species are paramagnetic

$O_2, S_2, O_2^+, N_2^-, B_2, NO_2, C\ell O_2$

34)



(A)



(C)

$AgBr + Na_2S_2O_3 \longrightarrow \underset{\text{(White ppt)}}{Ag_2S_2O_3\downarrow} + NaBr$

$\downarrow H_2O/\Delta$

$\underset{\text{(Black)}}{Ag_2S\downarrow} + H_2SO_4$

(E) (F)

X = 4 ; Y = 0

PART-3 : MATHEMATICS

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35) $g(x) = x^2 + ax + 2, h(x) = x^2 + bx + c$

$f + g = 2x^2 + (a-3)x - 5$ → r, t

$f + g = 2x^2 + (b-3)x + c - 7$ → r, s

$g + h = 2x^2 + (a+b)x + c + 2$ → s, t

$r + t = \frac{3-a}{2}$ ..... (i)

$r + g = \frac{3-b}{2}$ ..... (ii)

$s + t = -\frac{(a+b)}{2}$ ..... (iii)

(i) - (iii)

$r - s = \frac{3-b}{2}$

$r = \frac{3}{2}$ & $t = \frac{-5}{3}, rs = \frac{c-7}{2}$

$st = \frac{c+2}{2}$

$c = h(0) = \frac{52}{19}$

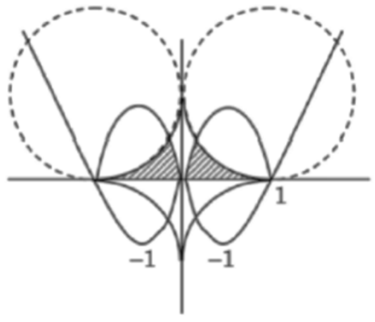
36) Volume will be independent of origin shifting $5|x| + 10|y| + |z| = 10$

v = 8[Volume of 5x + 10y + z = 10 in first octant]

$v = 8 \times \frac{1}{6}[1.2.10] = \frac{80}{3}$

37) Required area is

$$2\left[-\int_0^{\alpha} ex \ln x \, dx + \int_1^{\alpha} \left(1 - \sqrt{1-(x-1)^2}\right) dx\right]$$



38) $\sqrt{3a+13} = x$, $\sqrt{3b+13} = y$, $\sqrt{3x+13} = z$

$(x+y+z)^2 \leq 3\sum x^2$ (R.M.S ≥ A.M)

$(x+y+z)^2 \leq 3\left(3\sum a + 39\right)$

$x + y + z \leq 12\sqrt{3}$

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39) $|2A^3| = 2^{12} \Rightarrow |A|^3 = 2^9 \Rightarrow |A| = 8$

$\Rightarrow (\alpha + \beta)^2(\alpha - \beta) = 8$

$(\alpha + \beta)^2 = 4$ & $\alpha - \beta = 2$

$\Rightarrow \alpha + \beta = \pm 2$ and $\alpha - \beta = 2$

$\Rightarrow \alpha = 2, \beta = 0$ or $\alpha = 0, \beta = -2$

40) $$f(x) = \int_0^{3\pi/4} \left| x - \frac{3\sin t}{\cos t + \sqrt{2}} \right| \left(\frac{1 + \sqrt{2}\cos t}{(\cos t + \sqrt{2})^2} \right) dt$$

$$f(x) = \int_0^1 |x - 3u|\, du \Rightarrow f(x) = \begin{cases} \frac{3}{2} - x & ; \ x \leqslant 0 \\ \frac{x^2}{3} - x + \frac{3}{2} & ; \ 0 \leqslant x \leqslant 3 \\ x - \frac{3}{2} & ; \ x \geqslant 3 \end{cases}$$

41) $$x^2 \frac{dy}{dx} = y^2 e^{\frac{1}{x}} \Rightarrow \frac{dy}{y^2} = \frac{e^{\frac{1}{x}}}{x^2} dx$$

Integrating both sides,

$$\frac{-1}{y} = \int \frac{e^{\frac{1}{x}}}{x^2} dx + C$$

$\Rightarrow$ Putting $\frac{1}{x} = t \Rightarrow \frac{-1}{x^2} dx = dt \Rightarrow -\frac{1}{y} - \int e^t dt + C \Rightarrow$

$$\frac{-1}{y} = e^{-} + C \Rightarrow \frac{-1}{y} = -e^{\frac{1}{x}} + C$$

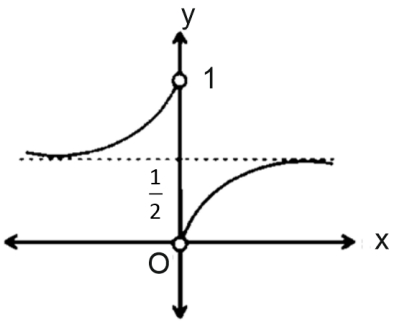
$\because \lim_{x \to 0^-} f(x) = 1 \Rightarrow -1 = 0 + C \Rightarrow C = -1$

$$\frac{-1}{y} = -e^{\frac{1}{x}} - 1 \Rightarrow y = \frac{1}{1 + e^{\frac{1}{x}}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\left(1 + e^{\frac{1}{x}}\right)^2} e^{\frac{1}{x}} \left(\frac{-1}{x^2} \right) = \frac{e^{\frac{1}{x}}}{x^2 \left(1 + e^{\frac{1}{x}}\right)^2}$$

$\because \frac{dy}{dx} > 0 \ \forall x \in R - \{0\}$

$\lim_{x \to \pm\infty} \frac{1}{1 + e^{\frac{1}{x}}} = \frac{1}{2}$ and $\lim_{x \to 0^+} \frac{1}{1 + e^{\frac{1}{x}}} = 0$



Graph of the function is From the graph option (A), (B) and (D) are correct.

42)

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We can determine the value of $P_r(L,N)$ as $= \dfrac{\text{coefficient of } x^L \text{ in } \left(x^1+x^2+x^3+.....x^r\right)^N}{r^N}$

The value of $P_3(5,2) = P_3(5,3) = \dfrac{2}{9}$; $P_9(15,2) = \dfrac{4}{81}$

$P_6(14,3) = P_6(14,5) = \dfrac{5}{72}$; $P_3(10,4) = P_3(10,6) = \dfrac{10}{81}$

$P_4(9,3) = P_4(9,4) = \dfrac{5}{32}$

43) We can determine the value of $P_r(L,N)$ as $= \dfrac{\text{coefficient of } x^L \text{ in } \left(x^1+x^2+x^3+.....x^r\right)^N}{r^N}$

The value of $P_3(5,2) = P_3(5,3) = \dfrac{2}{9}$; $P_9(15,2) = \dfrac{4}{81}$

$P_6(14,3) = P_6(14,5) = \dfrac{5}{72}$; $P_3(10,4) = P_3(10,6) = \dfrac{10}{81}$

$P_4(9,3) = P_4(9,4) = \dfrac{5}{32}$

44) $\text{A. adj}\left(A^2\right) = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$\Rightarrow |A|^2 I = A \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ ..... (1)

$\therefore$ A. adj $\left(A^2\right) = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ now determinant both sides

$|A| . \left|A^2\right|^2 = -1$

$\Rightarrow |A| = -1 \Rightarrow I = A \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$\Rightarrow B = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$\therefore B^2 - 3B^2 + B(1) + I(1) = 0$

$\Rightarrow B^{-1} = A = -B^2 + 3B - I$

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45) $$A.\,adj\left(A^2\right)=\begin{bmatrix}1&0&2\\2&1&0\\1&0&1\end{bmatrix}$$

$$\Rightarrow |A|^2 I = A\begin{bmatrix}1&0&2\\2&1&0\\1&0&1\end{bmatrix} \quad \ldots\ldots\ldots (1)$$

$$\therefore A.\,adj\left(A^2\right)=\begin{bmatrix}1&0&2\\2&1&0\\1&0&1\end{bmatrix}$$ now determinant both sides

$$|A|.\left|A^2\right|^2=-1$$

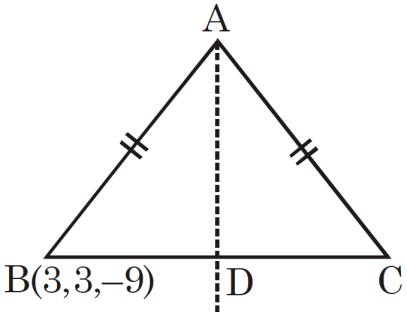
$$\Rightarrow |A|=-1 \Rightarrow I = A\begin{bmatrix}1&0&2\\2&1&0\\1&0&1\end{bmatrix}$$

$$\Rightarrow B=\begin{bmatrix}1&0&2\\2&1&0\\1&0&1\end{bmatrix}$$

$$\therefore B^2-3B^2+B(1)+I(1)=0$$

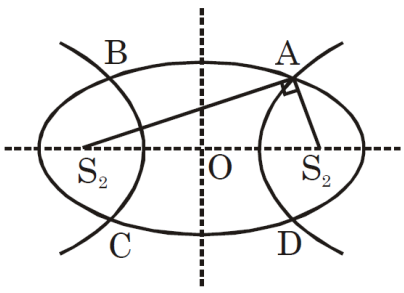
$$\Rightarrow B^{-1}=A=-B^2+3B-I$$

46) Let D is $(\lambda+1, 2\lambda+2, 3\lambda-3)$



$$\overrightarrow{BD}.(\hat{i}+2\hat{j}+3\hat{k})=0 \Rightarrow \lambda=-1$$

- D is mid point of BC
- C is (-3, -3, -3)



47)

$AS_1 + AS_2 = 2a = 10$

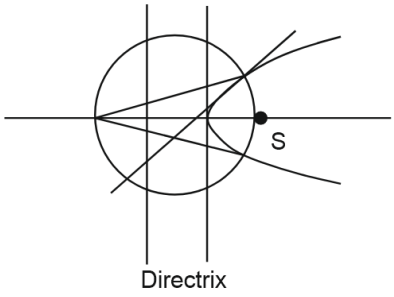
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$AS_2 - AS_1 = 2A = 6$

$AS_2 = 8$ & $AS_1 = 2$ & $S_1S_2 = \sqrt{68}$

for ellipse $2ae_1 = 2\sqrt{17} \Rightarrow e_1 = \frac{\sqrt{17}}{5}$

for hyperbola $2Ae_2 = 2\sqrt{17} \Rightarrow e_2 = \frac{\sqrt{17}}{3}$



48)

All lines are tangent to parabola $y^2 = 4x$

$x_1 = -1$ (Property of parabola)

$x_3 = 1$

49) Let $n^2 = 2^{62} \cdot 3^{38}$

Let F is a factor or $n^2 \Rightarrow 2^{\alpha}3^{\beta}$, $0 \leqslant \alpha < 62$; $0 \leqslant \beta \leqslant 38$

$\Rightarrow$ $n(F) = 6339 = 2457$

Let a, b are two factor, of n^2 such that a. b = n^2 when $a, b \neq n$

If a < n then b > n will exist in pairs

Number of factors less than n = 1228

Similarly if x and y are factors of n such that x. y = n and x and y are also factors of n^2

$\Rightarrow$ number of such factors = (32 × 20 –1) = 639

Req number of factors = 1228 – 639 = 589

A	B	A+B
$\bar{x}_1 = 40$	$\bar{x}_2 = 55$	$\bar{x} = 50$
$\sigma_1 = \alpha$	$\sigma_2 = 30 - \alpha$	$\sigma^2 = 350$
$n_1 = 100$	$n_2 = n$	$100 + n$

$$\bar{x} = \frac{100 \times 40 + 55n}{100 + n}$$

$$5000 + 50n = 4000 + 55n$$

$$1000 = 5n$$

50) $n = 200$

$$\sigma_1^2 = \frac{\sum x_i^2}{100} - 40^2$$

$$\sigma_2^2 = \frac{\sum x_i^2}{100} - 55^2$$

$$350 = \sigma^2 = \frac{\sum x_i^2 + \sum x_j^2}{300} - (\bar{x})^2$$

$$350 = \frac{\left(1600 + \alpha^2\right) \times 100 + \left[(30 - \alpha)^2 + 3025\right] \times 200}{300} - (50)^2$$

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$2850 \times 3 = \alpha^2 + 2(30-\alpha)^2 + 1600 + 6050$
$8550 = \alpha^2 + 2(30-\alpha)^2 + 7650$
$= \alpha^2 + 2(30-\alpha)^2 = 900$
$= \alpha^2 - 40\alpha + 300 = 0$
$\alpha = 10, 30$
$\sigma_1^2 + \sigma_2^2 = 10^2 + 20^2 = 500$

51) $8\left[\vec{a}\times\vec{b}\ \vec{b}\times\vec{c}\ \vec{c}\times\vec{a}\right] = 800$
$\Rightarrow \quad [\vec{a}\,\vec{b}\,\vec{c}]^2 = 100$
$\vec{b}.\left(\vec{a}\times\left(\vec{b}\times\left(\vec{c}\times\vec{b}\right)\right)\right)$
$= \vec{b}.\left(\vec{a}\times\left(\left|\vec{b}\right|^2\vec{c} - -\left(\vec{b}.\vec{c}\right)\vec{b}\right)\right)$
$= \left|\vec{b}\right|^2\left[\vec{b}\,\vec{a}\,\vec{c}\right] - 0$
$\Rightarrow \quad \left|\vec{b}\right|^2 = 25$
$\left|\vec{b}\right| = 5$

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Answer Key / Solution for this test is NOT AVAILABLE

(This document was not present in the source files.)